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MASTER

Design, fabrication and characterization of a Mach-Zehnder interferometric switch within the POLarization based Integration Scheme (POLIS)

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TTE-OED

Design, fabrication and characterization of a Mach-Zehnder interferometric switch within the POLarization based Integration Scheme (POLIS)

A.J.G.M. van de Hulsbeek May 2006

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Design, fabrication and characterization of a Mach-Zehnder Interferometric switch within the POLarization based Integration Scheme (POLIS)

Aert van de Hulsbeek

May 24, 2006

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1

INTRODUCTION

Information is knowledge, knowledge is progress. Where other inhabitants of our planet simply strive to survive and reproduce, modern man has evolved from this and developed the constant urge to make life more interesting and challenging. We are constantly looking for limits and trying to cross these limits, often by the use of technology.

In coming this far, information has played a key role, and perhaps more importantly, the sharing of information. Ever since the wheel was invented, the following generations could think of ways to put this amazing discovery to good use. In this way, by building on existing information, new inventions are possible.

New ways of communication allow us to learn even faster and use information to generate new ideas, new inventions and new challenges at an increasingly higher rate. This is where the challenge for the electrical engineer at the Opto-Electronic Devices group of the Eindhoven University of Technology lies: to think of ways to meet the demands of and desire for information.

1.1. OPTICAL COMMUNICATION

Transporting information from one point to another is one thing, but being able to process this information in the required way offers a totally different challenge. When we take the evolution in electronic communications as an example, we can see two major transitions: the realization of the telegraph, and that of the transistor.

While in 1830 the American Joseph Henry already demonstrated the use of the electromagnet for long distance electronic communication, it took until 1947 before a start could be made to electronically process the information over these lines. The invention of the transistor by John Bardeen, Walter Brittain and William Shockley played a key role in this evolution.

In optical communication, a similar trend can be observed. With the invention of the laser in 1960 in combination with a low-loss fiber created in 1970, the era of optical communications had started. A lot of effort is now put into finding ways to process this data in an all-optical fashion but also combining the advantages of electronic and optical devices. These efforts should lead to ways to utilize the tremendous amount of bandwidth the existing optical communication lines offer us.

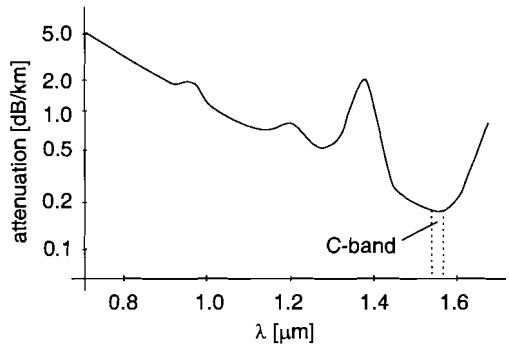


FIGURE 1.1: *Attenuation of an optical fiber currently in use, at various wavelengths.*

This bandwidth is exploited using techniques like wavelength division multiplexing (WDM), where each signal is assigned to its own wavelength. The wavelength region of main interest for optical communication is 1525 to 1565 nm (the C-band), where fiber losses are low and an erbium doped fiber amplifier (EDFA) can amplify the signal. See figure 1.1 for an illustration of this telecommunication window.

1.2. OPTOELECTRONIC INTEGRATED CIRCUITS

From the beginning of the optical communication era, the emphasis has shifted from 'further and faster' to 'more and cheaper'. Integration of optoelectronic devices focuses on reducing manufacturing and operating costs.

Advances in processing has allowed us to combine gain elements, detection elements, guiding structures, gratings, splitters and combiners, phase modulators and beam focusing elements into structures and devices. The main driving force behind this has been the performance requirements of optical communication networks.

While silicon has been the material of choice for integration of electronic functions, it is expected that III-V semiconductor based technologies are the best approach for optical devices, due to their ability to generate light efficiently as well as providing a full range of other required photonic functions.

The InGaAsP material system offers tuning of the bandgap in a broad range of interest within the C-band. By varying composition, the absorbing properties for certain wavelengths can easily be tuned, while keeping the material lattice-matched to the InP substrate.

Monolithic integration in this material system, which is achieved by combining active (devices which involve carrier recombination and generation, such as LEDs, lasers and detectors) and passive devices (such as waveguides and switches), overcomes interconnect issues posed by hybrid integration solutions. Furthermore, it opens the door to increased product volumes at reduced cost, since circuit sizes can be minimized and the robustness increases.

The processes utilized for monolithic integration include quantum well intermixing, butt-joined epitaxy, selective area epitaxy, ion implantation, dopant diffusion and many others.

1.3. POLIS OPPORTUNITIES

One of the major issues in making an integrated device is combining active and passive elements. Whereas active elements require absorption (recombination/generation effects) at a certain wavelength, transparency is required for passive elements. Therefore integration usually requires various materials with different absorption band edges to be grown on one wafer.

Due to technological limitations, it has always been difficult to reliably have different materials in different regions of a wafer. Device integration technology has therefore primarily focused on designs which allow the use of similar materials for all devices, which can be processed to change the material properties in spatially controlled segments of the wafer.

Although some of the proposed integration techniques offer ways to remove the need for a regrowth step (e.g. bandgap tuning by intermixing), the limited spatial selectivity puts serious boundaries on its usefulness.

The ultimate achievement for monolithic integration would be to find a technique which allows a full range of active and passive devices to be integrated on one wafer, by defining them during lithography.

This is exactly what the POLarization based Integration Scheme (POLIS) [1] has to offer. By exploiting the different material behavior for transverse electrical (TE) and transverse magnetic (TM) polarized light, and defining this polarization in different regions of the wafer in the design stage, active and passive optical functions can be realized without the need for a regrowth step. More details about the POLIS technique will be given in chapter 4

The main drawback of this technique is that the optical signals coming from a fiber have an undefined polarization and are therefore incompatible with POLIS. A way to deal with this is to employ a polarization splitter on the incoming signal, operate on both polarizations in a similar fashion in the optical chip and recombine them. This would require the chip to be fabricated in duplicate.

1.4. ELECTRO-OPTICAL SWITCHES

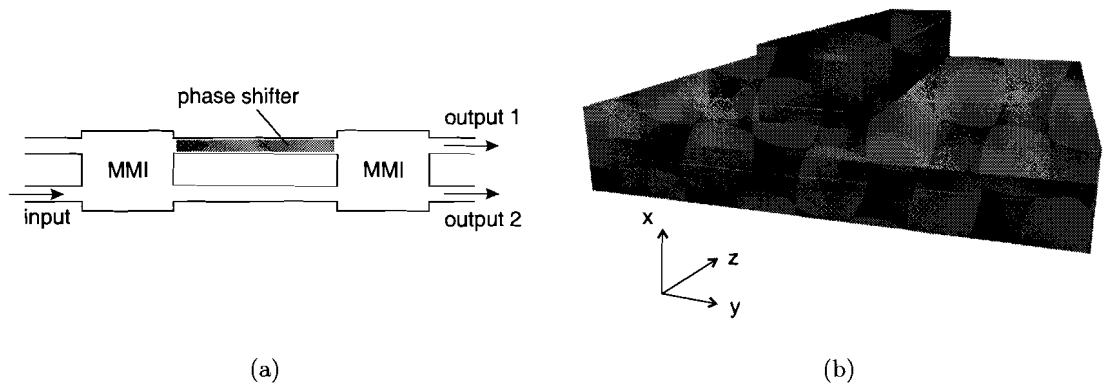
One of the key components in integrated optical circuits is the electro-optical switch (EOS). This device uses an externally generated electric field to change the path followed by a confined light wave.

EOS's have two major applications. The first is the generation of optical data, by modulation of a continuous optical signal. The second is the routing of signals in an optical network. At present the switching still takes place in the electrical domain, although it is believed that in the future a transfer to all-optical switching will be made for these routing networks.

The requirements for an EOS include the following:

- Low attenuation. In the EOS, part of the light will be absorbed by the material. It is important that the signal losses are minimized, to guarantee high performance operation of the integrated circuit as a whole.

- Low crosstalk. Different optical paths might interfere with each other to introduce



(a) (b)

FIGURE 1.2: *Layout of an MZI based switch (a) and directions as defined in this report (b).*

crosstalk. This can lead to serious degradation of the optical signal quality and therefore the performance, especially in large optical networks. The EOS should therefore introduce a minimal amount of crosstalk.

- Small dimensions. The advantages of a small footprint include lower propagation losses, easier integration and cost reduction. The size should not exceed a few millimeter in length.

- Low switching voltage. In general, it is easier to drive a high-frequency circuit at low voltages and currents to maintain the desired switching characteristics. The EOS should be able to operate at switching voltages of 5 V or less.

- Wavelength flexibility. The switch should be able to operate over wavelengths in the 1525 to 1565 nm range for communication applications.

- High switching speed. Packet-switched networks will require switches which have configuration times in the order of nanoseconds. The EOS should therefore be able to operate in the GHz range.

The polarization of the signal at a certain point in an optical network is generally not defined. Because of this, EOS's are required to have similar switching characteristics for both polarizations. However, for the signals within a POLIS circuit the polarization is always defined. Our EOS does therefore not have to meet this restriction.

A wide variety of electro-optical switches has been designed in the past, all with their own advantages and disadvantages. Most of these designs are fabrication intolerant (directional coupler EOS) or require large currents for operation (total internal reflection EOS, digital optical switch).

A Mach-Zehnder interferometer (MZI) based EOS does not have these disadvantages. It can operate under reverse bias, and the multi-mode interferometer (MMI) couplers used in the design are fabrication tolerant. In figure 1.2(a), a schematic representation of the MZI EOS is given. It features an MMI coupler which acts as a 3 dB power splitter, followed by two waveguides and another MMI coupler which acts as a combiner element.

1.5. Conventions

In this report, we will use a number of conventions for clarity.

- Directions. In a semiconductor context, the z-direction is usually used to indicate the growth direction, while in a device context the z-direction is used to indicate the direction of propagation. As the work presented in this report is done within a device framework, we will adapt the latter convention: x-direction is the direction of growth, the y-direction is the direction perpendicular to the direction of growth and propagation and the z-direction is the direction of propagation. These directions are clarified in figure 1.2(b), where the light propagates in the direction of the ridge.
- Units. All units are in S.I. units, unless indicated otherwise.
- Temperature. All values, measurement environments etc. mentioned in this report are at room temperature (for convenience set to 300 K).
- Vectors. Vectors will be written in bold typesetting, e.g. $\mathbf{k}$.

1.6. Outline of the report

The work presented here covers theory as well as practical work. First, some theoretical details will be described in chapter 2 for reference and a better understanding of various physical phenomena and device operation. In chapter 3, some background on the more practical issues is given. In chapter 4 more background on POLIS will be given and we discuss the work that was done in Australia, where the POLIS wafers that were used to realize the EOS's were made. Chapter 5 deals with the modeling of the phase shifter of the switch, and gives some more insight into various refractive index change mechanisms and the predicted switching efficiency. In chapter 6 the realization and characterization of the switch is described and its performance is discussed. Finally we end with recommendations and conclusions in chapter 7.

2

Theoretical considerations

2.1. Introduction

In this chapter, some theoretical background is treated which we need for the calculations and modeling in later chapters. First the relevant semiconductor properties are treated followed by a short description of a few optical concepts. Finally the operation of the most relevant devices is discussed.

2.2. Semiconductor basics

Crystal lattice

Under the correct growth conditions, semiconductors form a crystalline solid. In such a solid, all the atoms bond with each other in a regular pattern so that they are positioned on a periodic array of points in space. The crystal itself can therefore be described using a lattice and a unit cell. This unit cell is repeated at every point of the crystal lattice, and in this way describes the whole crystal. In figure 2.1 an example of a zinc blende unit cell is given, which is the crystal structure of all InGaAsP compositions.

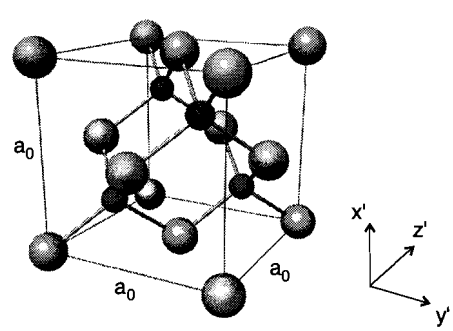


Figure 2.1: *A zinc blende unit cell.*

In this specific case, we can represent the unit cell as a cube (as we are working with a cubic crystal system). We introduce a cubic coordinate system and can therefore label the edges of the unit cell x', y' and z' which all have length a_0 under unperturbed conditions.

Each material has its own lattice constant a_0 and during epitaxy, the respective lattice constants should match within certain limits. If this is not the case, strain results. Small amounts of strain ($\leq 1\%$) when growing thin layers of material will not be a problem, as the lattice of the material adapts itself to that of the underlying material. If the lattice mismatch increases further, dislocations will start to form as the lattice is not able to adapt itself. These defects are highly unwanted, because they greatly enhance the optical losses in the material.

The amount of strain is determined by the the lattice constants of the two materials that are grown on each other. When the substrate and epitaxial layer have lattice constants $a_{0\text{sub}}$ and $a_{0\text{epi}}$ respectively and assuming the epitaxial layer will adapt itself to the substrate lattice, the strain $\varepsilon_{\parallel}$ in the directions parallel to the interface is given by:

$$\varepsilon_{\parallel} = \frac{a_{0\text{sub}} - a_{0\text{epi}}}{a_{0\text{epi}}} \tag{2.1}$$

where compressive strain is defined to be negative.

When the lattice is deformed in the in-plane directions, this also has an influence on the lattice constant in the growth direction and the resulting strain $\varepsilon_{\perp}$ is given by

$$\varepsilon_{\perp} = -2\frac{c_{12}}{c_{11}}\varepsilon_{\parallel} \tag{2.2}$$

where c_{12} and c_{11} are the elastic stiffness constants of the epitaxial material.

Direct semiconductors

When applying a current to a semiconductor, electrons can be injected into the conduction band of the material, and removed from the valence band. In doing this, electron-hole (e-h) pairs are created which can recombine resulting in the emission of a photon. This view is similar to that of an excited atom returning to its ground state.

For this transition, the usual laws of conservation of energy and momentum apply. For conservation of energy, the emitted photon will have an energy E_{ph} that is equal to the energy of the electron in the conduction band (initial state) minus the energy of the electron in the valence band (end state). The minimum energy difference between the valence and conduction band is called the band gap energy E_g. Under normal conditions, electrons mainly occupy the bottom of the conduction band E_c, and holes the top of the valence band E_v. Therefore the emitted photon will have an energy equal to or slightly larger than the band gap energy of the material. The corresponding wavelength of the photon is given by:

$$\lambda_0 = \frac{hc}{E_{ph}} = \frac{hc}{E_g} = \frac{hc}{E_c - E_v} \tag{2.3}$$

where h is Planck's constant and c is the velocity of light in a vacuum. For convenience, this relation is often written as

$$\lambda_0 = \frac{1.24}{E_{ph}}\,[\mu\text{m}] \tag{2.4}$$

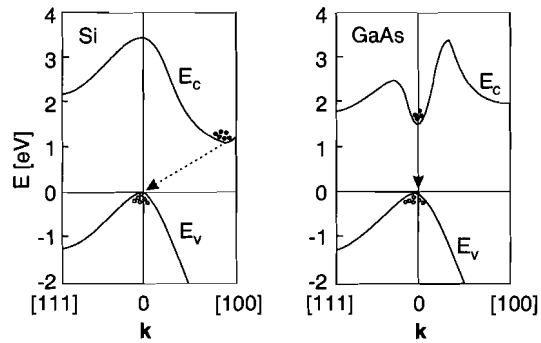


FIGURE 2.2: *Band structure of silicon (left) and galliumarsenide (right).*

with E_{ph} in eV.

The momentum of a photon is virtually zero in comparison to that of an electron. Because of this, the momentum of the electron in the initial (conduction band) state and end (valence band) state must be the same for a transition to take place. If this is not the case, an interaction with the crystal is needed to supply the missing momentum and make the transition possible. This lattice vibration is called a phonon.

The chance of a required phonon being available to make the transition possible is unlikely. Transitions that require no interaction with the crystal are therefore much more likely. If the required crystal interaction is zero for a transition between the minimum of the conduction band and the maximum of the valence band, we call the semiconductor direct. Figure 2.2 gives an example of an indirect (Si) and a direct (GaAs) semiconductor, for two crystal momentum directions.

As mentioned in the introduction, III-V semiconductor materials are the choice for photonic circuits, because of the direct bandgap, allowing light to be generated efficiently. This kind of undisturbed carrier recombination occurs with a high probability in these materials, with almost all injected carriers contributing to light generation. This phenomenon is utilized in devices like lasers and LED's. The same is true for the generation of an e-h pair by absorption of a photon. This effect is used for detectors.

THE FREE ELECTRON

In the classical picture, electrons are seen as particles having mass, position and speed. To be able to describe their behavior in semiconductors, a quantum mechanical treatment that considers the wave-particle duality of the electron is needed. The full wave function for a free particle, i.e. a plane wave, is given by $\Psi(\mathbf{k},\mathbf{r},t) = Ae^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}$ where $\mathbf{k}$ is the wave vector, $\mathbf{r}$ is the position, t is time, A is a constant denoting the amplitude and ω is the angular frequency.

We now only consider the spatial part of the wave function

$$\psi(\mathbf{k},\mathbf{r}) = Ae^{-i\mathbf{k}\cdot\mathbf{r}} \tag{2.5}$$

The length of $\mathbf{k}$ is called the wavenumber $k = |\mathbf{k}|$ and is related to the wavelength of the particle through $k = 2\pi/\lambda$. Noting that the momentum of the electron is directly related to

its wave vector through $\mathbf{p} = \hbar\mathbf{k}$, we can rewrite the energy of the free electron as follows

$$E(\mathbf{k}) = \frac{1}{2}m_0v^2 = \frac{p^2}{2m_0} = \frac{\hbar^2k^2}{2m_0} \tag{2.6}$$

where m_0 is the free electron rest mass. For a free electron, $E(\mathbf{k})$ is described by a parabola with the free electron mass m_0 as a parameter in all directions.

Effective mass approximation

When electrons move through a semiconductor, they are exposed to the periodic lattice potential of atoms in the crystal. As with all periodic structures, not all traveling waves can propagate through the lattice formed by the crystal: the waves can also be reflected and diffracted. Therefore not all $\mathbf{k}$ values are allowed in the semiconductor. Assuming a one dimensional periodicity a_x in the crystal, we get (in one dimension) constructive interference by reflection for $2a_x = n\lambda$. Using $\lambda = 2\pi/k$, we can write

$$k_x = \frac{n\pi}{a_x} \tag{2.7}$$

where n is an arbitrary integer value. This is simply the condition for a Bragg reflection. By extending this theory to three dimensions, it becomes clear that for a propagating electron wave, certain values of $\mathbf{k}$ are not allowed.

Away from these critical $\mathbf{k}$ values, we can approximate the behavior of the electron in a semiconductor as being a free particle with an adapted mass, representing the influence of the crystal potential lattice. This mass is called the effective electron mass m_e and is material dependent. We can therefore describe the valence band using the effective mass approximation:

$$E_c(\mathbf{k}) = \frac{\hbar^2k^2}{2m_e} \tag{2.8}$$

where the bottom of the conduction band ($\mathbf{k} = 0$) is set at $E_c = 0$. Because the free electron describes an s-like[1] orbital in the semiconductor, its effective mass is the same in all directions.

In typical optoelectronic materials, the minimum energy difference between the valence and the conduction band occurs at the $\mathbf{k} = 0$ point. An example of this is GaAs, as shown in figure 2.2. As carrier transitions mainly occur at this point, where the E-$\mathbf{k}$ relation is parabolic, the effective mass approximation allows for important simplifications in semiconductor theory.

Heavy and light holes

In a semiconductor context, a hole can be described as a missing electron in the valence band. Unlike an electron at the bottom of the conduction band, which has an s-like orbital (with zero angular momentum), holes describe a linear combination of p-like orbitals (with nonzero angular momentum). Three independent hole states can be distinguished, namely a heavy-hole state, a light-hole state and a split-off state all forming their own valence band. The split-off valence band is separated by the split-off energy, and is usually of less importance. It will be ignored in the rest of this report.

[1] This refers to the orbital of an electron in a hydrogenic atom

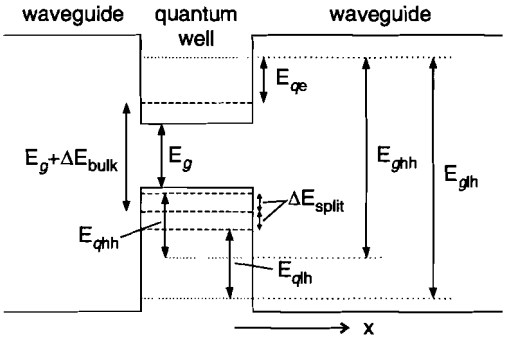


FIGURE 2.3: *Energy levels in a strained quantum well.*

In the effective mass approximation (assuming no band mixing), the valence bands are described by:

$$E_v(\mathbf{k}) = -E_g - \frac{\hbar^2 k^2}{2m_h(\mathbf{k})} \tag{2.9}$$

where E_v either describes the light hole band with effective mass m_{lh}, or the heavy hole band with effective mass m_{hh}. Unlike electrons in the conduction band, their effective mass depends on the crystal direction. Usually the effective mass in the principal directions (x, y and z) is considered.

STRAIN EFFECTS

Applying strain deforms the crystal lattice, and therefore the periodicity of the potential lattice through which the electron wave propagates. Because of this, the band structure of the material will change. In the following we will only consider a specific type of strain which occurs when materials with different lattice constants are grown on top of each other. This strain is called biaxial because its cause lies in the in-plane directions (y- and z directions). Because of this deformation there is also a resulting uniaxial perpendicular strain.

One of the implications of the change in periodicity of the crystal (due to the biaxial strain) is that the band gap of the bulk material changes. Also, because the (uniaxial) strain introduces an asymmetry in the crystal, the degeneracy of the light and heavy hole bands is removed, and both the e-lh and e-hh transition now feature their own band gap. These new band gaps are described by:

$$E_{glh} = E_g + \Delta E_{\text{bulk}} - \Delta E_{\text{split}} \tag{2.10}$$

$$E_{ghh} = E_g + \Delta E_{\text{bulk}} + \Delta E_{\text{split}} \tag{2.11}$$

where ΔE_{bulk} is the bulk band gap change and ΔE_{split} is the band splitting energy.

The biaxial hydrostatic component of the strain changes the lattice constant of the material and therefore introduces a change in the bulk band gap of the material:

$$\Delta E_{\text{bulk}} = -a(\varepsilon_\perp + 2\varepsilon_\parallel) \tag{2.12}$$

where a is the hydrostatic deformation potential. A compressive strain increases the band gap, while a tensile strain decreases it.

The uniaxial shear component disrupts the cubic symmetry of the semiconductor and breaks the degeneracy of the light and heavy hole band at the $\vec{k} = 0$ point. The splitting energy equals

$$\Delta E_{\text{split}} = -b(\varepsilon_{\parallel} - \varepsilon_{\perp}) \tag{2.13}$$

where b is the shear deformation potential. Compressive strain will move the light hole band down in energy by this amount and the heavy hole band up.

Quantum wells

Up to now we have been considering effects in bulk material. In these bulk materials, electrons and holes are free to move in any direction because their de Broglie wavelength is very small in comparison to the dimensions of the material. When the thickness of a semiconductor layer is reduced to a size comparable to this wavelength however, the motion of the carriers becomes restricted in the direction of growth. This restriction leads to a reduction in allowed energy states for the carriers, therefore quantizing the allowed energy states in this thin layer. This situation is similar to the well-known quantum mechanical treatment of a particle confined between potential barriers. If there is restriction in one direction, we speak of a quantum well.

In a physical situation, these potential barriers are formed by heterostructure junctions. The quantum well material has a lower band gap than the surrounding material, so that the carriers are trapped and assume a discrete energy state in the well. In appendix A the procedure to calculate these energy states is described. As the barrier height and effective mass is different for electrons, light holes and heavy holes, the confinement energies will also be different.

We can include the confinement effect in equations 2.10 and 2.11 by simply adding the quantization energies, which yields:

$$E_{glh} = E_g + \Delta E_{\text{bulk}} - \Delta E_{\text{split}} + E_{qlh} + E_{qe} \tag{2.14}$$

$$E_{ghh} = E_g + \Delta E_{\text{bulk}} + \Delta E_{\text{split}} + E_{qhh} + E_{qe} \tag{2.15}$$

where E_q denotes the quantization energy calculated from appendix A. These energies are illustrated in figure 2.3.

Quantum wells are currently the best choice for active semiconductor devices. This is mainly due to their step-like density of states profile (associated with the quantized levels), the possibility to tune emission wavelength without altering the material composition and ease of manufacturing.

2.3. Optical concepts

Refractive index

The refractive index n of a material is the factor by which electromagnetic radiation (like light) is slowed down as it travels through it, relative to propagation in a vacuum. The new speed becomes $v = \frac{c}{n}$ and the wavelength $\lambda = \frac{\lambda_0}{n}$, where c and λ_0 are the speed of light and the wavelength of the EM wave in a vacuum respectively.

The refractive index is given by:

$$n = \sqrt{\epsilon_r \mu_r} \tag{2.16}$$

where ϵ_r is the relative permittivity and μ_r is the relative permeability of the material. The permittivity of a material describes how it is affected by an electric field and vice versa, and is given by:

$$\epsilon = \epsilon_r \epsilon_0 \tag{2.17}$$

where ϵ_0 is the vacuum permittivity.

In general it holds that the denser a material is, the stronger the interaction between the material and the EM wave is. The amount of interaction is not only dependent on the material, but also on the frequency of the wave. This means that light with different wavelengths will experience a different refractive index when traveling through the same material. This is called dispersion.

The time-independent description of a traveling EM plane wave is given by equation 2.5, where ψ describes either the electric ($\mathbf{E}$) or magnetic ($\mathbf{H}$) field of the EM wave. For propagation in the z-direction of the electric field we can write:

$$\mathbf{E}(z) = E_0 e^{-ikz} = E_0 e^{-ink_0 z} \tag{2.18}$$

where E_0 is the amplitude of the electric wave, and the wavenumber k is given by $k = nk_0$, where $k_0 = 2\pi/\lambda_0$. A similar expression can be written for $\mathbf{H}$.

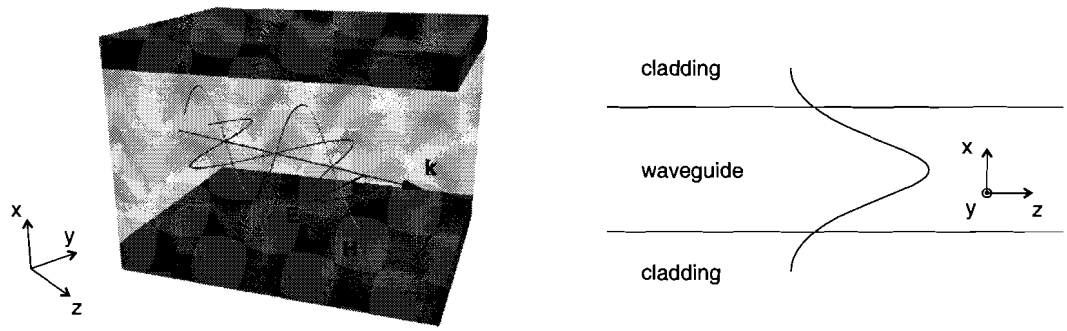
The propagation of an EM wave in a medium is effectively a constant interaction of the wave with the material: the electric field is the impulse and the material properties form the systems response. In this view the electron clouds in the medium are resonant at the frequency of the EM wave, absorbing the incident energy and transmitting it at the same frequency. This effectively slows down the EM wave as it propagates, which can be represented in equation 2.18 by having a real refractive index n larger than 1. If this energy is not retransmitted by the charge clouds, it is lost (e.g. in the form of heat) and contributes to the absorption of the material.

Losses in a medium can be described by an imaginary component of the refractive index, so that the total refractive index is described by a complex number $n = n' + in''$. From equation 2.18, we can see that the imaginary part of the refractive index represents a decaying exponential. The amplitude damping coefficient is defined as $\alpha_\mathrm{A} = -n''k_0$.

Considering only the electric field, the measurable quantity of the wave is the intensity $I(z) = \mathbf{E}(z)\mathbf{E}(z)^*$. The intensity in a lossy medium will therefore decay exponentially with $\alpha = 2\alpha_\mathrm{A}$. If α is given per centimeter, we can calculate the losses in dB/cm according to:

$$\alpha_{\mathrm{dB/cm}} = -10\log\left(\frac{I(z)}{I(0)}\right) = 10\log\left(\frac{E_0^2 e^{\alpha}}{E_0^2}\right) = 10\log(e^{\alpha}) = 10\log(e)\alpha \tag{2.19}$$

It is known from Snell's law that an EM wave traveling from one medium to another having a different refractive index will experience refraction. When it is propagating within the medium with a higher refractive index, it can experience total reflection and be contained within it. This principle is used in waveguides.



(a) (b)

FIGURE 2.4: *A TE polarized wave propagating in a medium (a) and a waveguide with an optical field distribution (b).*

POLARIZATION

Figure 2.4(a) shows a light wave traveling in a medium with direction **k**. The energy of the light is contained in an electric (**E**) and magnetic (**H**) wave[2], which both have their direction perpendicular to the direction of propagation. The orientation of these waves describes the polarization of the light.

As we are considering non-magnetic materials, the interaction of light with the crystal has an electric nature. Because of this, usually only the (orientation of the) electric field vector is considered. Two important cases can be distinguished when the light is traveling parallel to an interface, as shown figure 2.4(a). In the transverse electric (TE) case, **E** is oriented parallel to the interface, whereas for TM **H** is parallel.

The origin for these two cases lies in the solution of Maxwell's equations. For an anisotropic medium (where the relative permittivity ϵ_r is a tensor), two solutions to these can be found for propagation in an arbitrary direction. These solutions correspond to light propagating at different velocities but in the same direction. It can be shown that the two solutions are mutually orthogonal linear polarizations. In a bulk material like InP there is only one solution, but when anisotropy is introduced (e.g. by applying an electric field) or when ϵ_r becomes position-dependent (e.g. by having a stack of different materials), this changes.

If the light does not have a pure TE or TM polarization, the electric wave can be decomposed into a parallel and perpendicular part. If these components are in phase, the polarization is said to be linear. If this is not the case, the polarization is elliptical.

[2]Note that for a propagating wave, **E** and **H** are in phase, whereas they are out of phase by $\frac{\pi}{2}$ for a standing wave.

2.4. Basic optical device operation

Waveguide

Waveguides are the most basic and important devices in an integrated optical chip. They are used to transport light with low losses. Their operation relies on the fact that light can experience total reflection when it hits the boundary to a lower index material, thus containing it within the higher index material. This confinement is wanted in all directions except for that of propagation.

In the growth direction this is achieved by growing materials with different refractive indices on top of each other. The material with a higher refractive index (in our case $In_xGa_{1-x}As_yP_{1-y}$) is called the waveguide layer. The surrounding layer with a lower refractive index is called the cladding layer (here InP). These layers are illustrated in figure 2.4(b) together with an optical field distribution.

The optical field in this slab waveguide can be considered to be a standing wave pattern (sine-like) in the direction perpendicular to the interfaces (x-direction) and traveling along the waveguide parallel to the interfaces (z-direction). The possible realizations of the standing wave pattern are called the modes of the waveguide, and can be found by solving the dispersion relations as presented in appendix B. Which modes are supported by the waveguide depends on the refractive index difference at the interface, the thickness of the waveguide and the polarization.

A plane wave in a waveguide never travels fully parallel to the interfaces and when viewed as a ray, the light can be seen to bounce up and down between the top and bottom interface of a (slab) waveguide as it propagates through it. This ray makes a certain angle with the interface and according to Snell's law, only a certain range of angles is allowed. From the dispersion relations in appendix B, the possible angles can be calculated. This angle results in a TM mode having a small parallel **E** component and the TE mode having a small parallel **H** component.

Most of the optical power is contained within the waveguide, but part of it penetrates into the cladding layers. According to this, the effective index that the mode experiences is a combination of the refractive indices of the waveguide and cladding layers. Therefore each mode has its own effective index. As shown appendix B, the penetration depth is different for TE and TM and thus they will 'feel' a slightly different effective index. Thus the speed at which they travel in a medium can vary. This behavior gives rise to problems in designing polarization insensitive phase dependent devices.

The index contrast in other directions than the growth direction is achieved by removing material, which is called etching. It leaves a ridge for the light to propagate in, much like depicted in figure 1.2(b). If etching is done through the waveguide layer, it is called a deep waveguide and its contrast is formed between the waveguide and air. When this is not the case, the contrast is formed by the difference in effective index of the unetched and etched regions. The modes in these directions can be found in a similar manner as is done for slab waveguides.

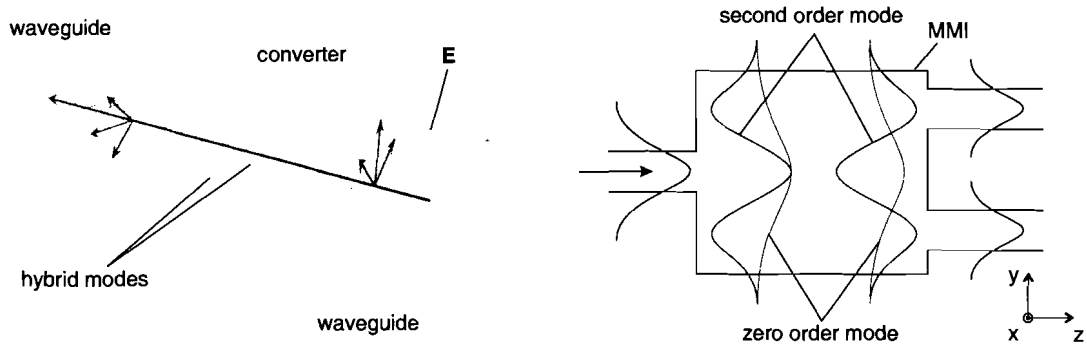


FIGURE 2.5: *Illustration of the principle of operation of a polarization converter (a) and of a 1x2 MMI power splitter (b).*

POLARIZATION CONVERTER

In a polarization converter, the optical axes are tilted under a certain angle as opposed to those in a normal waveguide. The electric field of a mode is decomposed into two components perpendicular to each other, which are called hybrid modes. These two components both experience a different effective index and will therefore be alternatively in- and out of phase. After propagation over a certain distance, the two components can be recombined in a waveguide again forming a different polarization.

There are different ways to tilt the optical axes, one of which is a slanted rib waveguide converter featuring one slanted sidewall. The components of the electrical field of the light traveling through it, will make an angle with the vertical sidewalls, determined by the width of the polarization converter. In the ideal case, the angle is 45° and both hybrid modes are excited equally so that full conversion can be obtained.

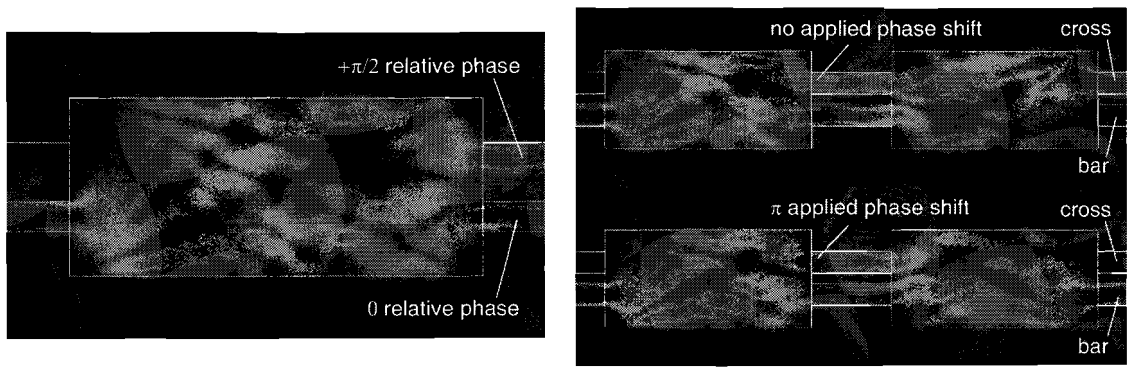
Figure 2.5(a) illustrates the conversion effect. Assuming the light entering the polarization converter is TM polarized, it will be converted into elliptical polarized light as the two components of the electric field become out of phase. After propagation over a longer distance the phase difference between the perpendicular components will have increased even further and they will be out of phase by π. If we end the polarization converter at this length, the components will recombine to form a TE polarization in a waveguide. If the length of the converter is doubled, we will end up with a TM polarization again.

It can easily be seen that TE-TM conversion length is equal to:

$$L_\pi = \frac{\lambda_0}{2|n_2 - n_1|} \tag{2.20}$$

where L_π is the converter length required to obtain full conversion and n_1 and n_2 are the effective indices of the hybrid modes. To minimize the converter length, the two components should have a large difference in effective index.

This description of a polarization converter is a very simple one, in reality a lot of non-idealities play a role in the behavior of the actual device [2]. Although a very important



(a) (b)

FIGURE 2.6: *BPM simulations of a restriced interference MMI (a) and cross and bar states of an MZI employing two of these MMI's (b).*

device for POLIS applications, the converter is not described in more detail here, as it is not needed for the EOS.

MULTI-MODE INTERFEROMETER

A multi-mode interferometer (MMI) is basically a waveguide that supports a range of modes. The operation of the MMI relies upon the fact that the various modes in it propagate at different speeds. As a result of this, the field profile of the combined modes will vary as they propagate through the MMI. An extensive overview of the imaging possibilities of MMI's is given in reference [3].

To illustrate the operation, we consider a simple example. We take an MMI that supports 3 modes, given in figure 2.5(b). Because of symmetry reasons, a gaussian shaped field entering the MMI will only excite the zero and second order mode. These modes have a different effective index. As can be seen from this figure, after propagation over a certain length, the two modes will be out of phase by exactly π. A simple summation of the fields shows that the input field is split into two parts. If we couple the two parts into separate waveguides at this position, we obtain a power splitting function.

The imaging capabilities of an MMI allow for various power splitting and combining functions. When restrictions are made on the excitement of the modes, it is called restricted interference. A particular case of this is called paired interference, which can yield a very short 2x2 MMI coupler. In figure 2.6(a), a beam propagation method (BPM) simulation of such a device is given. The field enters from the bottom input waveguide, and leaves through both output waveguides. The phase difference of $\frac{\pi}{2}$ of the fields in both output ports plays a key role in the operation of this device; other splitting/combining functions of the device can be deduced by considering symmetry and reciprocity.

MACH-ZEHNDER INTERFEROMETER

A Mach-Zehnder interferometer is a device that utilizes the constructive and destructive interference properties of light, by employing the phase information of the light wave. In general, the light is split into two parts, with a well-known phase relation. By adjusting the phase relation between these two parts and recombining them, we can either arrive at constructive or destructive interference.

A Mach-Zehnder interferometeric electro-optical switch is a device capable of spatially switching an input signal by introducing a phase change. A schematic view of an MZI EOS is given in figure 1.2(a). In our case, a 2x2 MMI coupler is used to split the input field into two waveguides. If no phase change in the waveguides takes place and a similar MMI is used to recombine the field, this will happen in the opposite waveguide (cross). When a phase shift of π is applied in one of the arms, the field will recombine in the other output waveguide (bar). This effect is illustrated in figure 2.6(b). The design presented here forms the basis of the EOS. The most important demands for such an EOS already have been described in section 1.4.

The phase shift is obtained by introducing a difference in refractive index of the material. This can be done in several ways and by means of several mechanisms. These will be described in chapter 5. Appendix C gives the theoretical behavior of a MZI as a function of an applied phase shift.

3

Experimental considerations

3.1. Introduction

An important part of the work described in this report has a practical nature. In this chapter we describe the techniques and equipment used to realize and characterize the wafers and the devices.

3.2. Fabrication

Metal-organic chemical vapor deposition growth

All samples described in this report were grown by metal-organic chemical vapor deposition (MOCVD) at the Australian National University. It is a method of epitaxial growth where a vapor mixture of metal-organics and hydrides containing the desired elements is passed over a heated substrate. Once the mixture reaches the heated cell, it decompose into radicals and a reaction takes place at the surface of the substrate. This results in the deposition of the desired material. To make the growth as uniform as possible, the wafer is constantly rotated.

The group III metal-organic precursors used for this work are trimethylgallium ($Ga(CH_3)_3$) and tri-methylindium ($In(CH_3)_3$). An ultra-pure carrier gas (H_2) is passed through the bubblers containing these compounds. The group hydride V precursors are supplied as arsine (AsH_3) and phosphine (PH_3) gases. Dietyhlzinc ($Zn(C_2H_5)_2$) and silane (SiH_4) are used as p- and n-type dopant precursors respectively. The overal chemical reaction that takes place is:

$$x\mathrm{In}(\mathrm{CH_3})_3 + (1-x)\mathrm{Ga}(\mathrm{CH_3})_3 + y\mathrm{AsH_3} + (1-y)\mathrm{PH_3} \rightarrow \mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}_y\mathrm{P}_{1-y} + 3\mathrm{CH_4} \quad (3.1)$$

Group V gases are supplied in excess to ensure that the growth rate is controlled by the availability of the group III elements. The growth rate is furthermore influenced by temperature and pressure. In the case of the growth of ternary and quaternary materials, the exact proportion of the gases is crucial. The growth temperature can also have a large impact on the composition, due to differences in bond activation energies. Therefore pressure and temperature are kept constant as much as possible during growth. A similar reasoning can be

applied to the dopants: the incorporation is controlled by the amount of gas, the temperature and growth rate. Usually many growth runs are needed to optimize the growth parameters.

Lithography

The term lithography is used to describe the steps required to get a desired pattern on a wafer, most commonly using light. It starts with the deposition of photoresist on the wafer. This material changes its properties under the influence of light. The resist is distributed by quickly rotating the sample. A short pre-bake follows to remove solvent remnants. The next step is the defining of the patterns by exposure to a light source. A mask aligner is used for this purpose, which projects the pattern of a mask onto the sample. The mask consists of a quartz plate with chrome patterning.

After exposure, the resist is developed. In the case of positive resist the lit portions are removed, whereas for negative resist the unlit portions get dissolved. A post-bake slightly modifies the morphology of the resist and causes it to be hardened. After this, the sample can be etched.

Wet chemical etching

After the lithography, the semiconductor material can be removed by a wet chemical etch directly to define device structures using the remaining photoresist as a masking material. Wet chemical etching is only used when the definition does not play a critical role.

The sample is placed in an acidic solution, which slowly dissolves the material. The advantage of a wet-chemical etch is that it can be very selective, i.e. it only etches certain semiconductor materials while leaving others unaffected. In general etch stop layers are included in the layer stack. These etch stops utilize the selectiveness of the etchant and make sure the etch depth is well defined.

Most etchants have a preferred plane of the crystal on which the etching stops, giving rise to slanted wall profiles. Underetch effects (material is removed from under the resist) can also make processing more difficult.

Reactive ion etching

In the reactive ion etching (RIE) process, the remaining photoresist after lithography can not be used as a masking material, as it gets removed in the RIE. The photoresist pattern is therefore first transferred into silicon nitride (SiN_x), or the photoresist is protected by a thin layer of titanium.

In the RIE, a plasma of charged particles is formed above the sample by a high frequency field. The created species react with the surface of the sample. Because the surface on which the sample is placed is biased, the ions in the plasma are attracted and accelerated. These hit the atoms at the surface, removing them. This etch process results in highly anisotropic vertical etch profiles. In our reactor CH_4/H_2 is used for the plasma. Polymers are formed that have to be removed after the etch.

The etch speed can be very well controlled, but varies for different InGaAsP compositions. The main drawback is the sidewall roughness that results from the process, which can lead to high waveguide losses. Smoothing steps can remove this roughness.

Plasma enhanced chemical vapor deposition

The SiN_x needed for the RIE process is deposited by plasma enhanced chemical vapor deposition (PECVD). Precursor gases containing the desired elements are introduced into a chamber and a plasma is created by a high frequency ionizing field. The radicals that are created react with eachother, forming the required compound on the sample surface.

Plasma stripping

In plasma stripping, the sample is placed in a chamber with an oxygen plasma. The plasma removes remnants of photoresist and polymers that result from the RIE. It also oxidizes the surface of the semiconductor, up to a few nanometers in depth. This oxide can then be removed by H_3PO_4, effectively smoothing and cleaning the semiconductor surface.

Electron beam evaporation

This process is used to deposit metal on a sample. The sample is placed upside down in a chamber above a crucible containing the metal. A filament generates a beam of electrons which is accelerated and magnetically steered to hit the crucible. The material in the crucible is evaporated and condenses inside the chamber, leaving a thin film on the sample. A quartz crystal is used to monitor the deposited thickness. The deposition rate can be controlled by adjusting the filament current.

3.3. Characterization

Scanning electron microscopy

In scanning electron microscopy (SEM), electrons originating from a filament are focused on the sample surface. The electron beam is then scanned across the surface. These primary electrons interact with the atoms in the sample, and an image is generated from electrons that are re-emitted from the sample surface. When a primary electron hits the sample surface, it will penetrate over a finite distance before hitting either an electron or nucleus of the atoms in the sample.

When it hits an electron, the primary electron will give off part of its energy, making it possible to ionize the atom. The released electrons usually have a low energy and are only detected if they originate from a small region around the beam spot. The resolution that can be obtained is very high, and for the images presented in this report in the order of 10 nm. The intensity of the image at a certain position is determined by the amount of electrons reaching the detector. As different materials and compositions have different work functions (the energy needed to free an electron from the surface), a slight contrast difference between

the compositions can usually be seen. This contrast can be greatly enhanced by a short stain etch, making a depth profile.

If the primary electron hits the nucleus, there will be an elastic collision. The backscattered electron will maintain most of its energy and for this reason can originate from a much larger area around the beam spot and sample depth. For this reason the resolution of the backscattered electron image is in the order of 1 μm. The number of backscattered electrons produced increases with increasing atomic number of the sample, giving a clear contrast for materials of different atomic composition.

X-RAY CRYSTALLOGRAPHY

X-ray crystallography can be used to precisely determine the lattice constant of a material. The sample is bombarded with X-rays, which interact with the material according to the spacing between the atoms and the angle of propagation. Reflections from the atomic planes will result in constructive or destructive interference.

The angle at which constructive interference is observed is directly related to the spacing and thus the lattice constant. By calibrating the angle of a certain reference material and comparing it with another material, it becomes clear whether the material is grown lattice matched or if strain is present due to a lattice mismatch.

ELECTROCHEMICAL CAPACITANCE-VOLTAGE

An electrochemical capacitance-voltage (ECV) profiler can be used to determine carrier concentrations as a function of depth in semiconductors. Using an electrolyte solution, a Schottky contact is formed between the semiconductor and the solution. In applying a reverse bias, a depletion region is formed at the interface. This depletion region acts as a capacitor and by modulating the bias, the change in capacity relating to the doping level can be measured.

To increase the range of depths at which the doping profile can be determined, the electrolyte can also behave as an etchant at forward bias. The semiconductor materials will get oxidized and become soluble in the electrolyte. The required chemical reaction can only take place if there are sufficient holes at the interface between the semiconductor and electrolyte. In the case of a p-type semiconductor under forward bias, this is not a problem. In the case of an n-type semiconductor, the holes have to be supplied by generating electron-hole pairs with an external light source.

PHOTOLUMINESCENCE

Photoluminesence (PL) measurements are a straightforward way to determine the bandgap of a semiconductor. A laser source of an energy higher than the bandgap of the material is used to excite electron-hole pairs. These carriers will interact with the crystal and relax to the lowest energy in the valence and conduction band. From here they will recombine and emit a photon with an energy corresponding to the band gap. The luminescence from the semiconductor is then analyzed with a monochromator, capable of determining the luminescence intensity for a specific wavelength over a broad wavelength range.

When the luminescence is taken from the top of a sample, we can only collect TE polarized light. If it is taken from a facet of a sample containing a quantum well however, we can discriminate between the emitted TE and TM polarized light using a polarizer. As we have seen in chapter 2, the heavy and light hole band gaps can be at different energies, and these transitions couple differently with TE and TM polarized light. This version of photolumiscence can therefore reveal more information about both transitions.

Fabry-Perot loss

Fabry-Perot (FP) measurements gives a loss figure for light propagating in a waveguide. The light from a laser is coupled into a waveguide using microscope objectives, after being modulated by a chopper. The emerging light is projected onto a photodetector which is connected to a lock-in amplifier together with the chopper signal.

The input and output facets of the waveguide are highly reflective and form a Fabry-Perot resonant cavity in which only certain wavelengths 'fit'. By sweeping the narrow laser spectrum over a small range, fringes will appear in time due to constructive or destructive interference at different wavelengths. Together with the knowledge of the reflectivity of the facets, the modulation depth of these fringes gives a number for the waveguide loss, without knowledge of the absolute amount of light intensities. This measurement is only reliable for a total loss of 5 dB or less, as the fringes become more noisy and shallow for higher losses.

Transmission

When steady measurement environments are desirable (e.g. for comparison), the high wavelength selectivity of the Fabry-Perot cavity can give serious problems. Small fluctuations in temperature can greatly influence the output intensity as it slightly increases or decreases the cavity length and this in turn leads to either constructive or destructive interference. To overcome this problem, an EDFA can be used together with a filter, giving a broader spectrum than a laser. This 'smears out' the effect of the fringes and the output intensity will be much less susceptible to slight perturbations. Except for the light source, the transmission setup is the same as the Fabry-Perot setup.

With knowledge of the input and output power, a loss estimate can be made. It is however not easy to know exactely how much light is coupled into the device under measurement, which limits its accuracy. The range of measureable losses is not limited like it is for the Fabry-Perot method.

4

Design, realization and characterization of the POLIS wafers

4.1. Introduction

Integration of active and passive devices is still one of the major steps to overcome towards reliable and cheap production of photonic integrated circuits, operating in the communication window around 1550 nm. The POLarization based Integration Scheme (POLIS) offers a way to achieve this, by employing the difference in behavior of a strained quantum well for TE and TM polarized light.

As described in chapter 2, we can achieve a large band splitting between heavy- and light hole bands in a strained quantum well. Holes in both these bands couple differently to an electric field in the plane of the quantum well (y and z-directions) and that perpendicular to the quantum well (x-direction). The reason for this lies in the linear combination of p-like orbitals that make up the state of the holes. These orbitals have their orientation either in the x, y, or z direction. The heavy hole state is a combination of p_y and p_z orbitals, whereas the light hole state is a combination of all three. For this reason only a light hole will experience a varying electric field strength in the x-direction and couple with it. TM polarized light, which (in bulk material) has all of its electric field in the x-direction, will therefore only couple to light hole transitions. The actual transition strengths can be found to be [6]:

$$\alpha_{\mathrm{TM}} = \alpha_x = \alpha_{elh} \tag{4.1}$$

$$\alpha_{\mathrm{TE}} = \alpha_{y,z} = \frac{1}{4}\alpha_{elh} + \frac{3}{4}\alpha_{ehh} \tag{4.2}$$

We will see in the next section that for propagation in a waveguide with finite dimensions, a TM mode always has a small part of its electric field in the in-plane directions and therefore also slightly couples with the heavy hole transition.

If we now bring the e-hh bandgap energy down and the e-lh bandgap energy up as in a quantum well under compressive strain, TM polarized light will experience a larger bandgap than TE polarized light. For a specific wavelength (1550 nm) we can achieve high absorption

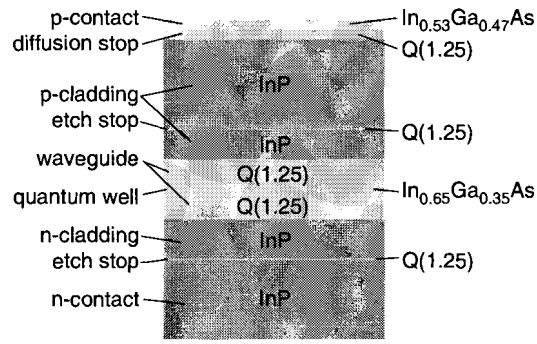


FIGURE 4.1: *A reference POLIS layer stack.*

for TE polarized light, and low absorption for TM polarized light. This is the first demand for the POLIS layer stack. Moreover, as the heavy hole state is the lowest energy state in the quantum well, active devices will mainly emit photons from this level. The second demand is therefore to have an e-hh bandgap of 0.8 eV (1550 nm).

Polarization converters, whose operation is described in section 2.4, are used to determine the polarization in a certain part of the chip. Passive devices will operate on TM polarized light, active devices on TE. So together with the specific behavior of POLIS material, the polarization converters can determine the function of the material without having to change it.

All material presented here was grown at the group of Electronic Materials Engineering at the Australian National University. In the past, this group has grown several POLIS wafers, but these did not give the desired behavior [4, 5]. Either the band splitting was insufficient, resulting in high losses for TM, or the emission wavelength was not at 1550 nm. This was enough reason to make another attempt at realizing a layer stack with proper POLIS behavior. The work in Australia mainly focused on the quantum well and waveguide thickness and composition. Several growth runs were done to meet the strict POLIS demands.

4.2. DESIGN CONSIDERATIONS

In this section the most important issues involved with the design of the layer stack are presented. First we turn to the influence of doping on the optical loss of the layer stack, followed by more background on the design of the material layers. We then discuss the design of the quantum well, responsible for the specific POLIS behavior and the influence of its position in the waveguide layer. A reference POLIS layer stack is given in figure 4.1.

DOPING LOSSES

Optical losses are one of the main considerations in the design of any wave guiding layer stack. These losses can have various origins, both on a material and a device level. The latter is not a material dependent parameter and is introduced during the fabrication processes, for example by RIE as described in section 3.2.

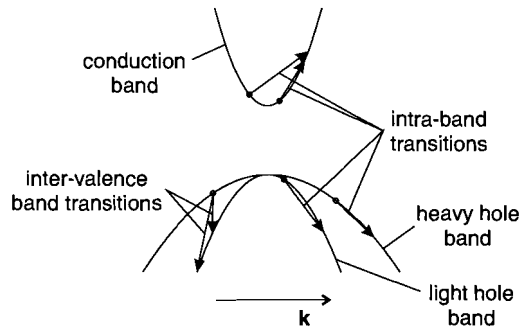


FIGURE 4.2: *Illustration of free-carrier absorption mechanisms.*

Loss mechanisms that are inherent to the material are band to band absorption, absorption due to crystal imperfections and free carrier absorption. The first can be minimized by simply having a sufficiently high bandgap. The second results in defect levels within the bandgap, and is also known as the Urbach tail of the material. This absorption tail stretches out into the bandgap of the material and allows for transitions below the bandgap energy. It should therefore be significantly higher than the operating photon energy to minimize these losses (e.g. 0.2 eV). For 1550 nm operation, this would imply a bandgap of approximately 1.0 eV.

The last source of material related loss is free carrier absorption. For this absorption, there are two possible mechanism, depicted in figure 4.2:

- A carrier moves to a higher energy state in the same band by absorbing a photon. This type of absorption can occur for both electrons and holes, and therefore for both p- and n-type materials. This is called intra-band absorption. Transitions that require less interaction with the lattice (change of $\mathbf{k}$), are more likely to occur. It can be seen from the shape of the bands that this type of transition will occur more easily for electrons than for holes.

- A carrier moves to another band by absorbing a photon. This type of absorption can only occur for holes, as there is only one conduction band, but multiple valence bands. It is therefore called inter-valence band absorption and only occurs in p-type materials. These transitions can take place even without interaction with the crystal, and are therefore much more likely to occur.

The intra-band absorption coefficient α_{ib} for a doping concentration χ due to this effect can be estimated by [20]:

$$\alpha_{\text{ib}} = \frac{e^3 \lambda_0^2}{10^7 \pi c} \frac{\chi}{m^2 \mu n} \tag{4.3}$$

where e is the electron charge, λ_0 is the photon wavelength in a vacuum, c is the speed of light in a vacuum, χ is the doping concentration (P or N), m is the carrier effective mass (m_h or m_e), μ is the carrier mobility (μ_h or μ_e) and n is the refractive index.

A more convenient equation (for $\lambda_0 = 1550$ nm) is given by:

$$\alpha_{\text{ib}} = 1.26 \cdot 10^{-16} \frac{\chi}{m^2 \mu_e n} \, [\text{cm}^{-1}] \tag{4.4}$$

where $\chi\,[\mathrm{cm}^{-3}]$ is the doping concentration (N or P), m $[m_0]$ is the carrier effective mass (m_e or m_h), $\mu\,[\mathrm{cm}^2\mathrm{V}^{-1}\mathrm{s}^{-1}]$ the carrier mobility (μ_e or μ_h) and n the refractive index. All these material and doping level dependent parameters can be calculated from appendix D and the hole density of states effective mass m_h, is given by [9]:

$$m_h = (m_{lh}^{3/2} + m_{hh}^{3/2})^{2/3} \tag{4.5}$$

Because the effective mass of holes is much larger than that of electrons, the intra-band losses are less for holes. Consider equation 4.4 and the material parameters from appendix D for InP. At a doping level of 10^{18} cm^{-3}, the losses due to intra-band absorption are predicted to be 10 dB/cm for n-doping, while for p-doping this is 3.8 dB/cm.

For p-type doped materials, inter-valence band absorption is the main contributor to doping loss. It is less material dependent and can be approximated by [10]:

$$\alpha_{P\mathrm{iv}} = 4.252 \cdot 10^{-20} e^{-3.657E} P \tag{4.6}$$

where $E\,[\mathrm{eV}]$ is the photon energy and P is the doping concentration. Taking $E = 0.8$, we can this simplify to:

$$\alpha_{P\mathrm{iv}} = 2.28 \cdot 10^{-17} P\,[\mathrm{cm}^{-1}] \tag{4.7}$$

now with P in cm^{-3}. A p-doping of 10^{18} cm^{-3} corresponds to a inter-valence band absorption loss of 99 dB/cm.

Both loss figures are valid for doping levels in the order of 10^{16} to 10^{19} cm^{-3}. In general, the losses due to n-type doping will be an order of magnitude lower than those due to he same level of p-doping. An overlap of the optical field with highly p-doped regions should therefore always be avoided in the design.

Waveguide layer

The design of the waveguide is mainly governed by optical loss considerations, consisting of losses in the waveguide itself as well as losses due to penetration of the optical field into the cladding layers. This penetration follows directly from equation 32 in appendix B, and is due to the limited refractive index contrast between the waveguide and cladding layer. Because the cladding layer is highly doped, it will have higher losses.

A higher index contrast (and therefore better confinement) will be achieved as the waveguide material approaches $\mathrm{In_{0.53}Ga_{0.47}As}$. The bandgap of this material however drops just as quickly, and the Urbach tail will start to play a role. As a result of this tail and the operating wavelength of 1550 nm, the bandgap of the material should be around 1.0 eV, so Q(1.25)[1] was chosen. The composition of this material lattice matched to InP is $\mathrm{In_{0.75}Ga_{0.25}As_{0.54}P_{0.46}}$. Due to the conflicting material demands for optical confinement and the influence of the Urbach tail, the growth should match this design as closely as possible.

The design for the width of the waveguide is also governed by the confinement. The wider the waveguide, the more light is contained within the waveguide. To avoid supporting multiple modes in this direction however, this width is limited to approximately 700 nm.

[1] Q(1.25) indicates an InGaAsP composition so that it has a bandgap corresponding to an emission wavelength of 1.25 μm.

Based on the limited growth data for quaternary materials at the ANU, a first test growth was done. Using PL (bandgap), X-ray (lattice constant) and SEM (thickness) measurements, the composition and thickness of the waveguide material were determined. Growth conditions were adjusted to better match the design, as required. A slight lattice mismatch (0.1%) for these materials is virtually inevitable, and the material is normally said to be lattice matched when below this value.

The difficulty in growing the exact lattice matched quaternary material is the fact that there are two group III elements, indium and gallium, to incorporate. Because of the high V/III ratio used during growth, the availability of these elements determines the growth rate. Hence, a slight fluctuation in the proportion of the indium and gallium flow will introduce a fluctuation in their solid fractions. It can therefore be expected that from growth to growth, the exact quaternary composition will vary.

Cladding, contact and etch stop layers

The dopants used in all cases were zinc for p-type and silicon for n-type doping. Zn has the well known issue of diffusion: measurements showed that a designed abrupt change in doping takes over 150 nm to drop by an order of magnitude. Care must therefore be taken when designing and using Zn as a dopant. To minimize losses, a gradual decrease in doping levels towards the waveguide layer is used.

An attempt was made to use carbon (C) as a p-type dopant. It has the advantage that its diffusion is much less than that of zinc, but unfortunately it does not behave as a p-type dopant for all InGaAsP compositions [7]. While not ideal, carbon doping could be used for the contact layer with the best reported [8] hole concentration of 6e19 cm^{-3} for InGaAs lattice matched to InP. This was not done however, as it is not the main purpose of this work.

The cladding layers consist of InP which has a lower refractive index than the waveguide and will give the desired optical confinement. Moreover it has a higher bandgap, so that carriers can easily reach the waveguide and quantum well layer. The thicknesses are designed so that the overlap of the optical field with highly absorbing layers is minimal, while keeping low-resistivity carrier transport towards the active region possible. The design of the cladding will be discussed further in the following sections.

The structure was grown on a semi-insulating substrate, which has the advantage of reduced electrical crosstalk between different devices on the same wafer. Both p- and n-contacts have to be fabricated on top of the structure. This requires making contacts at different depths, therefore various etch stop layers were added to the layer stack. To make good ohmic contacts, the top layer consists of highly p-doped InGaAs, and the bottom n-contact layer consists of highly n-doped InP.

Except for the issues with Zn diffusion, growing these layers is very straightforward. Thickness and composition problems should not play a role here as growth data is available for these materials.

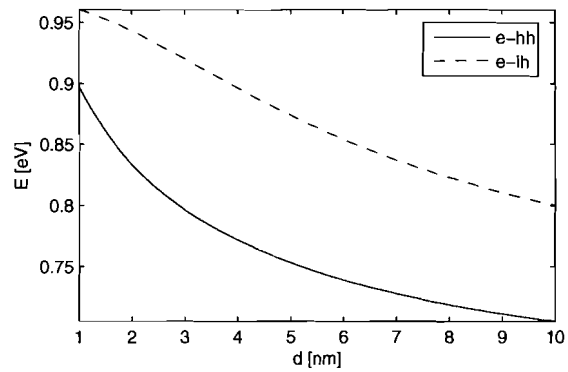


FIGURE 4.3: *The calculated e-hh and e-lh energies for different well widths.*

QUANTUM WELL LAYER

Using the procedure described in section 2.2, we can define an optimal design for the quantum well. The demands for the quantum well are an e-hh transition energy of 800 meV, and a high level of band splitting. Together with the composition of the waveguiding layer the transition energies can be calculated for varying quantum well compositions and widths. Figure 4.3 shows a plot of the e-hh and e-lh transition energies for varying widths of a $In_{0.65}Ga_{0.35}As$ quantum well with $In_{0.75}Ga_{0.25}As_{0.54}P_{0.46}$ surrounding material. From this figure it can be seen that a 3 nm $In_{0.65}Ga_{0.35}As$ quantum well is predicted to have an e-hh transition energy of about 0.8 eV (1550 nm) and a band splitting of about 120 meV. The exact numbers depend on the composition of the waveguide, but only vary a little.

Although the band splitting is quite large, a TM mode will experience some loss due to the Urbach tail of the e-lh bandgap. As mentioned for the waveguide layer, the operating photon energy should be in the order of 200 meV below that of the bandgap. This difference is less for the quantum well, but the optical overlap is not as large as for the waveguide material. The magnitude of these losses will be small, probably in the order of 1 dB/cm.

To realize the designed quantum well layer, a lot of effort was put into optimizing growth parameters. First, two samples with multiple quantum wells of different width were grown to check the confinement effect. Photoluminescence measurements were done to determine the e-hh bandgap energy of the wells. These were done at low temperature (7 K) to be able to resolve the multiple PL peaks, as the PL emission becomes very broad at room temperature. The measured energies were then corrected for temperature dependence.

Based on these results, the growth time was slightly decreased and the next growths aimed at a 3 nm well. In the three subsequent growths, the strain of the well material was increased while the designed thickness was kept constant. This shifted the e-hh emission peak towards 1550 nm as desired, while giving more bandgap energy splitting. The final design featured a 3 nm $In_{0.65}Ga_{0.35}As$ well, having 0.8% compressive strain. This design was used for all samples presented in this report. As mentioned in section 2.2, this amount of strain in a thin layer should not give any defects in the crystal, because it is well below the critical thickness.

As the quantum well is only a few tens of atomic layers thick, a slight variation in the thickness of this layer will have a large impact on the confinement energies and therefore the emission wavelength of the sample. Issues like these confirm the need for several test growths.

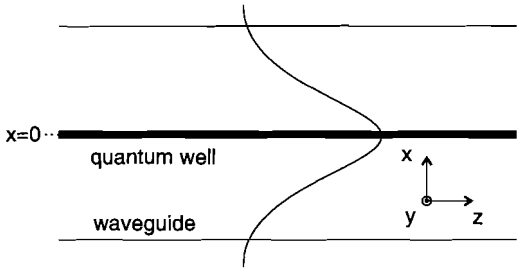


FIGURE 4.4: *Intensity distribution of the propagating part of a fundamental TE mode in a waveguide.*

POSITION OF THE QUANTUM WELL

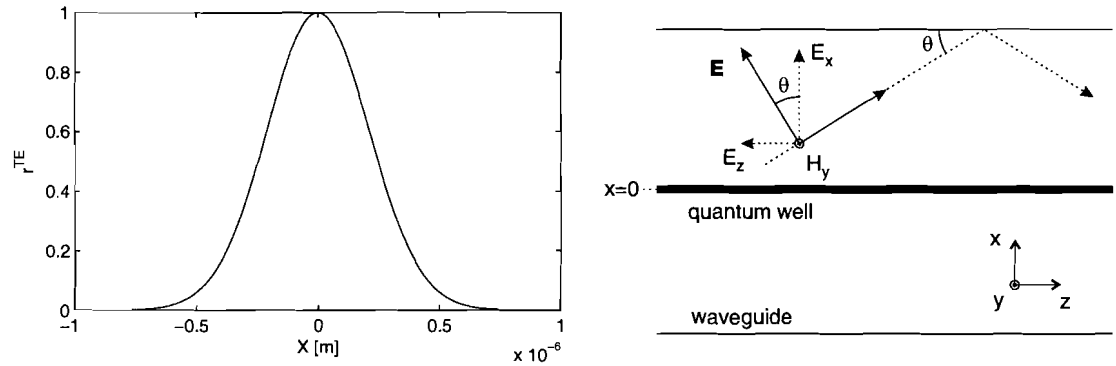
The quantum well is the active element, responsible for both absorption and generation of TE light in the structure. Normally it is placed in the middle, but its position can be varied within the waveguide.

Because holes have a lower mobility and the background doping of all undoped materials used is slightly n-type, they will have trouble reaching the quantum well through the undoped waveguide. The resistivity of the device will therefore be greatly influenced by the distance the holes to travel through the waveguide. If we place the quantum well closer to the top of the waveguide, holes will reach this active layer more easily. Moreover it allows for slightly increasing the n-doping of the waveguide layer under the quantum well. This will also improve the conductivity but most importantly change the switching behavior favorably, as we will discuss in chapter 5.

Conversely, it also results in a reduced absorption of the TE polarized light, which is undesirable. Consider the slab waveguide in figure 4.4. Because of the limited thickness of the waveguiding layer, it only supports one perpendicular fundamental mode (in the x-direction). The intensity distribution of the mode can be approximated by a gaussian, as described in appendix B. As we are dealing with a TE mode, the electric field is parallel to the plane of the quantum well. The overlap of the field with the absorbing quantum well will determine the absorption that the mode experiences. As the optical intensity is highest in the middle of the waveguide, TE absorption will be highest when it is placed in the middle of the waveguide. If it is placed out of the middle, the reduced overlap of the optical field with the quantum well will reduce the absorption.

The TE absorption for samples with a 3 nm quantum well in the middle of a 600 nm waveguide is known from previous experiments. Therefore, we can estimate the absorption at a different position by determining the ratio between these two values.

Setting the middle of the waveguide at $x = 0$, the ratio r^{TE} of the absorption of the quantum



(a) (b)

FIGURE 4.5: *TE absorption ratio for a thin quantum well at different positions in a 600 nm waveguide (a) and a ray representation of a TM wave propagating in a waveguide (b).*

well at position $x = X$ to that at position $x = 0$ for the fundamental TE mode is given by:

$$r^{\mathrm{TE}} = \frac{\alpha_{x=X}}{\alpha_{x=0}} = \frac{\int_{X-\frac{1}{2}d}^{X+\frac{1}{2}d} I(x)dx}{\int_{-\frac{1}{2}d}^{\frac{1}{2}d} I(x)dx} \tag{4.8}$$

where d is the thickness of the quantum well and $I(x)$ the intensity distribution of the TE mode in the x-direction.

For the TE mode, all electrical field power is carried by in-plane components of **E**. For very thin quantum wells, such as described here, and by using the gaussian distribution, we can simplify this ratio to:

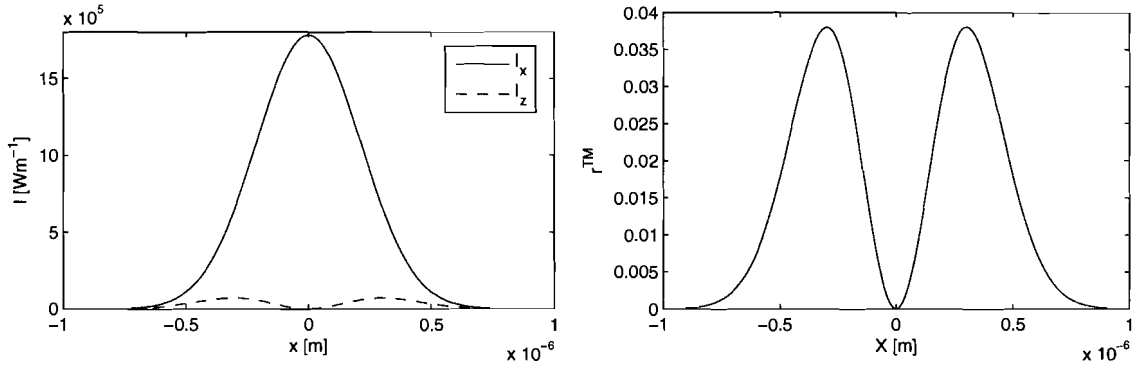
$$r^{\mathrm{TE}} \approx \frac{I(X)d}{I(0)d} \approx e^{\frac{-4\pi X^2}{d_{m\mathrm{eff}}^2}} \tag{4.9}$$

where we used the definition for $I(x)$ and $d_{m\mathrm{eff}}$ defined in appendix B. This ratio is illustrated in figure 4.5(a).

The influence of the position of the quantum well for TM polarized light is more complicated. Consider figure 4.5(b), where a ray of TM polarized light is traveling through a waveguide. Because of the POLIS material behavior, the majority of the electric field does not experience attenuation as it is perpendicular to the quantum well. However, the small E-field component E_z that arises due to the propagation angle θ does, as it is in the plane of the quantum well. The component E_z has a different distribution than the E_x component. A derivation of this fact, based on Maxwell's equations is given in appendix E. The distributions of both E-field components are illustrated in figure 3(a) of the same appendix.

From figure 3(a) it is clear that E_z is only zero in the middle of the (symmetric) waveguide, at $x = 0$. This implies that if the quantum well is placed at this position, the part of the light that is in this component will not be absorbed. When this is not the case, it will be attenuated by the quantum well, resulting in additional losses for the TM mode.

By solving the propagation angle θ as described in appendix B, we can calculate what portion of the optical power is contained in the E_x and E_z component. We use some straightforward



(a) (b)

FIGURE 4.6: *Intensity distribution of the x- and z-component of* **E** *(a) and the TM/TE absorption ratio (b).*

trigonometry using the ray approach. Again, consider figure 4.5(b) where a ray hits the boundary of waveguide and cladding under an angle θ. The amplitude of E_x and E_z can be seen to be:

$$E_x = E\cos\theta \tag{4.10}$$

$$E_z = E\sin\theta \tag{4.11}$$

where $E = |\mathbf{E}|$. The power in an electromagnetic wave component is proportional to its amplitude squared, and thus we can write:

$$P_x = E_x^2 = E^2\cos^2\theta \tag{4.12}$$

$$P_z = E_z^2 = E^2\sin^2\theta \tag{4.13}$$

Setting the total power to unity ($E^2 = 1$), and using the distribution of I_x and I_z given in appendix E, we can write

$$I_x(x) = \cos^2\theta\frac{2}{d_{m\text{eff}}}e^{-\frac{4\pi x^2}{d_{m\text{eff}}^2}} \tag{4.14}$$

$$I_z(x) = \sin^2\theta\frac{16\pi}{d_{m\text{eff}}^3}x^2e^{-\frac{4\pi x^2}{d_{m\text{eff}}^2}} \tag{4.15}$$

These distributions are illustrated in figure 4.6(a) for a 600 nm waveguide.

Similarly to the dependence of the TE absorption on the position of the quantum well, illustrated in equation 4.9, we can write the ratio of TM absorption for a quantum well at position $x = X$ over the TE absorption for a quantum well at position $x = 0$ as:

$$\frac{\alpha_{x=X}^{\text{TM}}}{\alpha_{x=0}^{\text{TE}}} = \frac{\int_{X-\frac{1}{2}d}^{X+\frac{1}{2}d} I_z^{\text{TM}}(x)dx}{\int_{-\frac{1}{2}d}^{\frac{1}{2}d} I^{\text{TE}}(x)dx} \approx \sin^2\theta\frac{8\pi}{d_{m\text{eff}}^2}X^2e^{-\frac{4\pi X^2}{d_{m\text{eff}}^2}} \tag{4.16}$$

An illustration of this ratio is given in figure 4.6(b). A thinner waveguide will enhance this ratio substantially, as the propagation angle θ of the fundamental mode increases and $d_{m\text{eff}}$ only increases slightly.

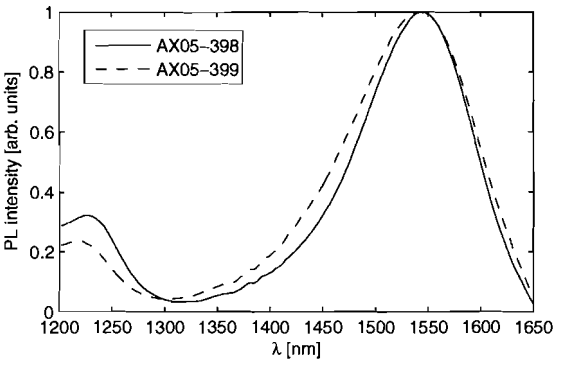


FIGURE 4.7: *Photoluminescence spectra of AX05-398 and AX05-399.*

		AX05-398					*AX05-399*				
Layer	Purpose	d (nm)	Material	Doping (cm^{-3})	n'	n''	d (nm)	Material	Doping (cm^{-3})	n'	n''
15	p-contact	100	InGaAs	P 1e19	3.540	-0.1	100	InGaAs	P 1e19	3.540	-0.1
14	diffusion stop	100	Q(1.25)	P 1e18	3.384	-2.92e-4	100	Q(1.25)	P 1e18	3.384	-2.92e-4
13	p-cladding	100	InP	P 1e18	3.166	-2.91e-4	100	InP	P 1e18	3.166	-2.91e-4
12	p-cladding	100	InP	P 5e17	3.166	-1.45e-4	100	InP	P 5e17	3.166	-1.45e-4
11	etch stop	20	Q(1.25)	P 5e17	3.384	-1.45e-4	20	Q(1.25)	P 5e17	3.384	-1.45e-4
10	p-cladding	250	InP	P 5e17	3.166	-1.45e-4	250	InP	P 5e17	3.166	-1.45e-4
9	etch stop	20	Q(1.25)	P 5e17	3.384	-1.45e-4	20	Q(1.25)	P 5e17	3.384	-1.45e-4
8	p-cladding	50	InP	I	3.166	-1.14e-7	250	InP	I	3.166	-1.14e-7
7	waveguide	300	Q(1.25)	I	3.384	-5.66e-8	100	Q(1.25)	I	3.384	-5.66e-8
6	quantum well	3	InGaAs	I	-	-	3	InGaAs	I	-	-
5	waveguide	300	Q(1.25)	I	3.384	-5.66e-8	500	Q(1.25)	N 3e16	3.384	-3.82e-7
4	n-cladding	5	InP	I	3.166	-1.14e-7	5	InP	N 5e17	3.166	-2.11e-5
3	n-cladding	500	InP	N 5e17	3.166	-2.11e-5	500	InP	N 5e17	3.166	-2.11e-5
2	etch stop	20	Q(1.25)	N 5e17	3.384	-8.76e-6	20	Q(1.25)	N 5e17	3.384	-8.76e-6
1	n-contact	500	InP	N 1e18	3.166	-5.03e-5	500	InP	N 1e18	3.166	-5.03e-5

TABLE 4.1: *Design values of AX05-398 and AX05-399. Background doping of intrinsic material is assumed to be n-type, 5e15* cm^{-3}*. Refractive indices are at* $\lambda_0 = 1550$ *nm.*

The derivation given here considers a slab waveguide, but it is also valid for channel waveguides. For these waveguides, an extra term comes into play because of the finite propagation angle in the y-z plane. The electric field component in the y-direction is then a consequence of the tilting of the resulting component in the z-direction that we derived earlier. This component is very small and will not have a significant contribution to the loss of the waveguide. It will therefore be neglected.

4.3. AX05-398 AND AX05-399

For this work two samples were first grown: *AX05-398* and *AX05-399*. In *AX05-398*, the quantum well was placed in the middle of the waveguide layer, 300 nm from the top cladding. In the case of *AX05-399* it was placed 100 nm from the top cladding. Furthermore in *AX05-399*, the waveguide layer under the quantum well was slightly n-doped.

Photoluminescence plots are given in figure 4.7. Both samples have their PL peak intensity slightly below 1550 nm. The growth of the quantum well is obviously well controlled between runs. The composition of the quaternary material can be seen to vary somewhat between *AX05-398* and *AX05-399*, but both have their peak around 1230 nm. This indicates that a composition close to Q(1.25) was obtained.

The layer stacks and most important material parameters are given in table 4.1. For the transparent materials, n' (the real part of the refractive index) is extracted from appendix D and n'' (the imaginary part) is determined by considering both the doping and equations 4.4 and 4.7. For simplicity, the doping is assumed not to have a substantial influence on n'. Typically, this introduces an error of at most 1e-3 for n'. For comparison, a 1% compositional deviation gives a 4e-3 error.

With a bandgap of 0.75 eV, the InGaAs contact layer (15) is not transparent at 1550 nm. Therefore its refractive indices can't be calculated in this manner. Instead, appendix D and reference [15] are consulted for the appropriate values. To be able to use equation 52 in appendix D, we choose a value of $\Delta E = -0.05$ eV instead of 0.05 eV. Reference [15] indicates that this is a valid approximation. The estimated absorption is 8e3 cm^{-1}, giving $n'' \approx -0.1$. For these high absorption values, the influence of doping on n'' is negligible. The InGaAs quantum well is not included in this calculation due to its limited thickness and low absorption for the TM polarization.

With these material parameters, we can use a complex mode solver to estimate the optical loss. We used the WASMF software tool, which calculates possible modes in the layer stack, similar to that calculated in appendix C. It is also able to solve for complex refractive indices, therefore not only giving the resulting effective indices but also a loss figure. WASMF is a 1D slab solver, so it is assumed that all of the optical field propagates in the layer stack as calculated by WASMF. In reality this is not the case, in particular for shallow waveguides. Tests show that this is not an important issue as 99% of the optical power travels directly underneath a ridge waveguide [36], and good estimates can still be made in all cases.

We first consider *AX05-398*, in three situations. The first is with the contact layer and top cladding removed, the second is with all layers intact (as given in table 4.1) and the final is with all layers including a metal contact. This contact consists of 50 nm titanium (Ti) and 500 nm gold (Au). From reference [11] we find their complex refractive indices at 1550 nm to be $n = 5.31 - 7.04i$ and $n = 0.18 - 10.2i$ respectively.

WASMF predicts the following losses:

- Layer 12-15 removed: WASMF: 3.3 dB/cm, measured 6-10 dB/cm. The additional losses are probably due to fabrication imperfections (scattering due to sidewall roughness), and a slight influence from the e-lh Urbach tail of the quantum well.
- All layers, no metal. WASMF: 18.8 dB/cm, measured: 18-23 dB/cm. These losses are due to the penetration of the optical field in the higher doped cladding layers.
- All layers, with metal: WASMF: 157 dB/cm, not measured. This excessive loss is mainly due to the combination of a high real and imaginary part of the refractive index of Ti.

It is thus obvious that when a metal contact is made on this material, it will result in unacceptable losses. The main reason for this is that the cladding layer (13) was made too thin. This is illustrated in figure 4.8, where the losses for *AX05-398* are plotted versus the thickness of layer 13. From this figure we can also see that the thickness of this layer should be at least 800 nm to minimize losses.

For *AX05-399* we can expect additional losses due to slight doping of the waveguide (layer

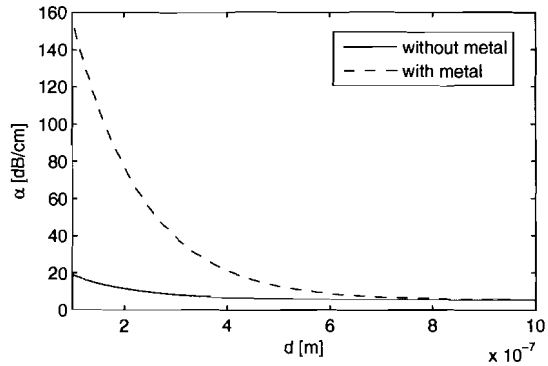


FIGURE 4.8: *Absorption versus the thickness of layer 13 of AX05-398, as calculated with WASMF.*

		AX06-029					AX06-030				
Layer	Purpose	d (nm)	Material	Doping (cm^{-3})	n'	n''	d (nm)	Material	Doping (cm^{-3})	n'	n''
15	protection	20	InP	I	3.166	-1.14e-7	100	InP	I	3.166	-1.14e-7
14	p-contact	100	InGaAs	P 1e19	3.540	-0.1	100	InGaAs	P 1e19	3.540	-0.1
13	diffusion stop	100	Q(1.25)	P 1e18	3.384	-2.92e-4	100	Q(1.25)	P 1e18	3.384	-2.92e-4
12	p-cladding	800	InP	P 1e18	3.166	-2.91e-4	800	InP	P 1e18	3.166	-2.91e-4
11	p-cladding	100	InP	P 5e17	3.166	-1.45e-4	100	InP	P 5e17	3.166	-1.45e-4
10	etch stop	20	Q(1.25)	P 5e17	3.384	-1.45e-4	20	Q(1.25)	P 5e17	3.384	-1.45e-4
9	p-cladding	250	InP	P 5e17	3.166	-1.45e-4	250	InP	P 5e17	3.166	-1.45e-4
8	p-cladding	50	InP	I	3.166	-1.14e-7	50	InP	I	3.166	-1.14e-7
7	waveguide	300	Q(1.25)	I	3.384	-5.66e-8	100	Q(1.25)	I	3.384	-5.66e-8
6	quantum well	3	InGaAs	I	-	-	3	InGaAs	I	-	-
5	waveguide	300	Q(1.25)	I	3.384	-5.66e-8	500	Q(1.25)	N 3e16	3.384	-3.82e-7
4	n-cladding	5	InP	I	3.166	-1.14e-7	5	InP	N 5e17	3.166	-2.11e-5
3	n-cladding	400	InP	N 5e17	3.166	-2.11e-5	400	InP	N 5e17	3.166	-2.11e-5
2	etch stop	20	Q(1.25)	N 5e17	3.384	-8.76e-6	20	Q(1.25)	N 5e17	3.384	-8.76e-6
1	n-contact	800	InP	N 1e18	3.166	-5.03e-5	800	InP	N 1e18	3.166	-5.03e-5

TABLE 4.2: *Design values of AX06-029 and AX06-030.*

5), in the order of 0.1 dB/cm. A bigger effect can be expected from the changed position of the quantum well. As explained in section 4.2, part of the light will get absorbed because the in-plane **E** component for TM is non-zero at positions other than the middle of the waveguide. For our 600 nm waveguide, the absorption ratio as given in equation 4.16 is calculated to be 0.0294. From measurements, the TE absorption is estimated to be in the order of 150 dB/cm. The corresponding TM absorption due to the quantum well displacement is therefore 4.4 dB/cm.

This leads to the assumption that losses will be in total 4.5 dB/cm higher for *AX05-399* than for *AX05-398*. This is confirmed by measurements: with contact layer and top cladding removed, the losses were measured to be 10-14 dB/cm. With these layers the losses are 23-25 dB/cm.

4.4. AX06-029 AND AX06-030

For the next growths, the issue with the thin cladding was solved by increasing the thickness to 800 nm. Furthermore some small modifications were made to make processing easier and an InP protection layer was added. The resulting layer stack design is given in table 4.2.

Before processing, the layer thicknesses were checked using a profilometer. The thickness measurements were done by iterations of a selective etch followed by a depth measurement

	AX06-029			AX06-030		
Layer(s)	Designed d (nm)	Measured d (nm)	Error	Designed d (nm)	Measured d (nm)	Error
13&14	200	192	-4%	200	184	-8%
11&12	900	768	-15%	900	760	-16%
8&9	300	253	-16%	300	246	-18%
5&7	600	641	+7%	600	654	+9%
3	400	-	-	400	331	-17%

TABLE 4.3: *Designed and measured thicknesses of the layers of AX06-029 and AX06-030. Thin layers are not considered.*

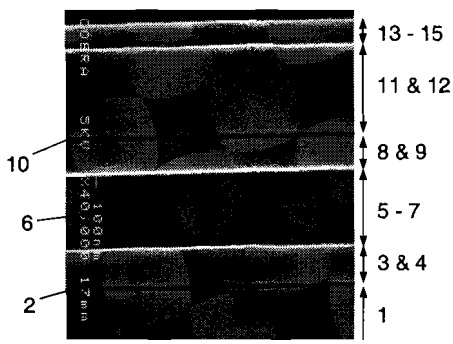


FIGURE 4.9: *SEM image of AX06-029 after a stain etch, layer numbers indicated.*

using the profilometer. The resulting values are given in table 4.3 and are averages taken from a few positions on the samples. It is very clear from this table that the InP is about 15% too thin, while the quaternary is about 8% too thick. Figure 4.9 gives a SEM image of *AX06-029* after a short stain etch.

The consequences of these variations are that losses will be higher than designed when a contact layer is present. WASMF calculations on the actual layer stack of *AX06-029* predict losses of 2.6 dB/cm for layers 11-14 removed, 4.9 dB/cm for all layers intact and 7.8 dB/cm with metal contacts. In all cases the protection layer (15) was not taken into account.

The actual measured values were much higher, approximately 17 dB/cm without contact layer and metal and 28 dB/cm with these for both samples. It is not fully clear what the origin of these high losses is. The POLIS behavior of the wafers has been verified during Fabry-Perot measurements and showed a very high absorption for TE. This indicates that the quantum well is intact, but the confinement energies might have shifted. SEM images taken from the realized waveguides show that the high TM losses are not caused by sidewall roughness. The most likely explanation is that of zinc diffusion towards the waveguide layer, due to the longer growth time of layer 12. The high losses and the resulting scatter of the measurement data disguised the difference in absorption between *AX06-029* and *AX06-030.*

The additional loss for waveguides with a contact layer and metal is also higher than expected. It was calculated to be 5.2 dB/cm, but measurements show values twice as large. Possible explanations are that the used refractive indices for the metal were not accurate for our deposition method, or that some diffusion of the metal into the semiconductor has taken place.

4.5. Conclusions

First, two POLIS wafers *AX05-398* and *AX05-399* were realized by careful design and consideration of the losses. The loss mechanisms were modeled and simulated and a very good agreement between the predicted values and the measured values was obtained. Unfortunately the top cladding was made too thin, so two new wafers were realized solving this issue.

These wafers, *AX06-029* and *AX06-030*, were used to realize the switches. They are almost identical, except for the position of the quantum well and the doping level of the waveguide layer under the quantum well. In the case of *AX06-029*, the quantum well layer is positioned in the middle of the waveguide and the waveguide is not intentionally doped. In *AX06-030* the quantum well was moved towards the top cladding and the waveguide layer under it was slightly doped. The origin of the additional losses measured for these wafers is not yet clear.

5

Modeling of the phase shifter

5.1. Introduction

The basis of the MZI switching is the change of refractive index. This mechanism introduces a phase difference in one of the arms of the MZI (the phase shifting section) and will effectively determine how efficient the switching behavior will be. Naturally, a switch with a small driving voltage and a small footprint is desirable. The phase shifter will determine these properties. We first model the useable electro-optical effects, and the light and electric field distribution in the phase shifter. Based on this, the effective phase shift at a certain bias can be calculated.

5.2. Absorption around the bandgap

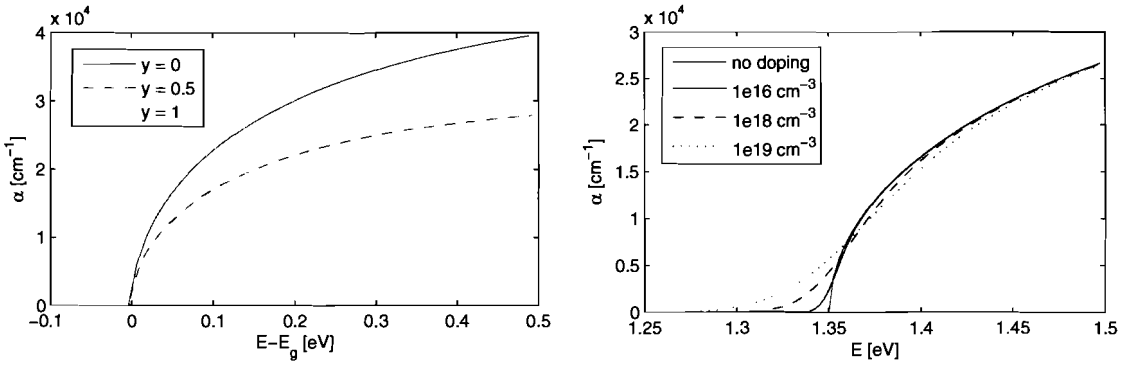
The bandgap shrinkage and bandfilling effects that will be discussed in the following section, impose a change of the bandgap. To be able to accurately describe the change in absorption because of these effects, we need a proper model. We apply the square-root law [22] for absorption near the bandgap. This model is valid in the parabolic band approximation, for energies close to the bandgap:

$$\begin{aligned} \alpha_g(E) &= \frac{C}{E}\sqrt{E - E_g} && E \geq E_g \\ \alpha_g(E) &= 0 && E < E_g \end{aligned} \tag{5.1}$$

where α_g is the intensity absorption coefficient near the bandgap, C is a fitting constant, E is the photon energy and E_g is the bandgap of the material under consideration.

We first turn to the magnitude of C, and determine it according to reference [23], using the absorption data from reference [15]. Knowing that the absorption due to doping is negligible at 0.2 eV above the bandgap, we interpolate the data in for different compositions of $InGaAs_yP_{1-y}$. For InP at this photon energy, the absorption is seen to be $3.0 \cdot 10^4$ cm^{-1}, and for $In_{0.53}Ga_{0.47}As$ this is estimated to be $1.4 \cdot 10^4$ cm^{-1}. So using equation 5.1 and these values, we can write:

$$\alpha_g(E_g + 0.2) = \frac{C}{E_g + 0.2}\sqrt{0.2} = 3 \cdot 10^4 - 1.6 \cdot 10^4 y \,[\text{cm}^{-1}] \tag{5.2}$$



(a) (b)

FIGURE 5.1: *Absorption for photon energies around the bandgap for different InGaAsP compositions (a) and effect of increasing doping levels according to Kane [16] in InP (b).*

with all energies in eV and C in $\text{eV}^{\frac{1}{2}}\text{cm}^{-1}$. Using the compositional dependence of E_g given in appendix D, this gives us the following description for C:

$$C = \frac{1.55 - 0.72y + 0.12y^2}{\sqrt{0.2}}(3 \cdot 10^4 - 1.6 \cdot 10^4 y)\ [\text{eV}^{\frac{1}{2}}\text{cm}^{-1}] \tag{5.3}$$

The absorption according to this model for energies less than 0.5 eV above the bandgap energy is given in figure 5.1(a) for different $\text{InGaAs}_y\text{P}_{1-y}$ compositions.

When a material is doped, there is a random distribution of ionized donors or acceptors present. These charged impurities result in potential fluctuations of the band edges, effectively making a tail that stretches out into the bandgap of the material. This tail is called the Urbach tail and was already briefly described in section 4.2. To treat the magnitude of this tail in more detail, we adopt the method of Kane [16] describing the band-edge energy fluctuations due to these impurities, given by:

$$\sigma = \frac{e^2}{4\pi\epsilon}\sqrt{2\pi\chi r_s} \tag{5.4}$$

where ϵ is the permittivity of the material, χ is the doping level and r_s the screening radius. For non-degenerate[1] systems, this screening radius is given by the Debye screening radius:

$$r_s = \sqrt{\frac{\epsilon kT}{e^2\chi}} \tag{5.5}$$

where k is Boltzmann's constant and T is the temperature (300 K). For degenerate doping levels it is given by the Thomas-Fermi screening radius:

$$r_s = \pi^{\frac{2}{3}}\sqrt{\frac{\epsilon\hbar^2}{e^2 m(3\chi)^{\frac{1}{3}}}} \tag{5.6}$$

[1]See appendix F for a definition of degeneracy

Effect	Origin	Occurs in	$\Delta\alpha/\Delta n$
Stark	field-induced	bulk material	$\Delta\alpha$
Franz-Keldysh	field-induced	bulk material	$\Delta\alpha$
Pockels (linear E.O.)	field-induced	bulk material	Δn
Kerr (quadratic E.O.)	field-induced	bulk material	Δn
quantum confined Stark (QCSE)	field-induced	quantum structures	$\Delta\alpha$
plasma	carrier	bulk material	$\Delta\alpha$
bandfilling (Burnstein-Moss)	carrier	bulk material	$\Delta\alpha$
bandgap shrinkage (renormalization)	carrier	bulk material	$\Delta\alpha$

TABLE 5.1: *Various descriptions of refractive index change effects, including their origin, the structure they occur in and the change they describe.*

where $\hbar$ is the reduced Planck's constant. In the case of p-type doping, m and χ are replaced by m_h and P respectively and for n-type doping by m_e and N.

For small energy fluctuations, we can adopt a Gaussian distribution. The probability that the absorption at energy E' is found at energy E is given by:

$$p\left(\alpha(E) = \alpha(E')\right) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{E'-E}{\sigma})^2} \tag{5.7}$$

Using this distribution, we can now integrate for all energy levels, effectively summing over all contributions of absorption at energy E. We deduce this approximation from the description given by equation 5.1 where we have an absorption below the bandgap of 0, so that we can set the lower integration limit to E_g:

$$\alpha_g(E) = \int_{E_g}^{\infty} \frac{C}{E'} \sqrt{E' - E_g} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{E'-E}{\sigma})^2} dE' \,[\mathrm{cm}^{-1}] \tag{5.8}$$

The doping dependency of the absorption edge is clearly illustrated in figure 5.1(b), where N was varied between 0 and 1e19 cm^{-3} in InP.

5.3. REFRACTIVE INDEX CHANGE MECHANISMS

Table 5.1 gives an overview of the relevant descriptions of the refractive index change mechanisms or effects. Except for the quantum confined Stark effect (QCSE), that is specific for quantum structures, all effects occur in bulk InGaAsP. Field-induced and carrier effects are distinguished, according to their origin.

Some effects are related to each other, but either emphasize the change in absorption of a material ($\Delta\alpha$)or focus on the change of refractive index (Δn). The Kramers-Kronig relation however prescribes that both are related, and that one can be derived from the other.

From table 5.1 we can see that only the Pockels and Kerr effect describe a change in refractive index directly, whereas for the other effects this change has to be derived through the Kramers-Kronig relation. We will first consider this relation.

The Kramers-Kronig relation

The Kramers-Kronig relation for an electromagnetic wave in a medium relates the real and imaginary part of the permittivity of that medium. The relation is derived in reference [17], where the wave-matter interaction is described as a forced oscillation of resonators formed by the electron clouds in the material. The perturbation is caused by the electric field of the light (as we are considering a non-magnetic medium), which oscillates with an angular frequency $\omega = \frac{E_{ph}}{\hbar}$. The Kramers-Kronig relation comes from a purely mathematical derivation but has serious physical consequences.

The relation is equivalent to a relation between refractive index and absorption of that medium. The latter is given by:

$$n(\omega) = 1 + \frac{c}{\pi}\mathcal{P}\int_0^\infty \frac{\alpha(\omega')}{\omega'^2 - \omega^2}d\omega' \tag{5.9}$$

where $\mathcal{P}$ denotes the principal value of the integral and α is the absorption coefficient. We normally consider photon energy rather than angular frequency, so we rewrite the equation:

$$n(E) = 1 + \frac{hc}{2\pi^2}\mathcal{P}\int_0^\infty \frac{\alpha(E')}{E'^2 - E^2}dE' \tag{5.10}$$

where E is the photon energy. This equation clearly shows that to be able to calculate the refractive index at a certain photon energy, the absorption spectrum for all energies has to be known.

Similarly, we can write the following for the change in refractive index as a consequence of a change in absorption:

$$\Delta n(E) = \frac{hc}{2\pi^2}\mathcal{P}\int_0^\infty \frac{\Delta\alpha(E')}{E'^2 - E^2}dE' = \mathcal{K}\{\Delta\alpha(E)\} \tag{5.11}$$

where $\mathcal{K}\{\}$ indicates this form of the Kramers-Kronig transform. This form replaces the required knowledge of the absorption over the whole energy spectrum, with the knowledge of the change in absorption over the whole spectrum.

The Stark effect

The Stark effect effectively changes the energy of electron states by influencing their orbits through the application of an electric field. It can be divided in two types: the linear and quadratic Stark effect. The first only occurs for atomic energy states higher than the ground state (principal quantum number > 1). In this case, there are multiple degenerate states having the same energy. Applying perturbation theory and selection rules, it can be shown that the degenerate states are separated from their original energy level in the opposite direction, by an amount proportional to the applied electric field strength [28]. All states however feature a quadratic Stark shift [29], which has a cubic dependence on the Bohr radius of the carrier and a quadratic dependence on the applied electric field strength.

A more quantitative description of the Stark effect will not be given here, as the effect in bulk InGaAsP is small. More importantly, the influence on the refractive index is described directly by the Pockels and Kerr effects.

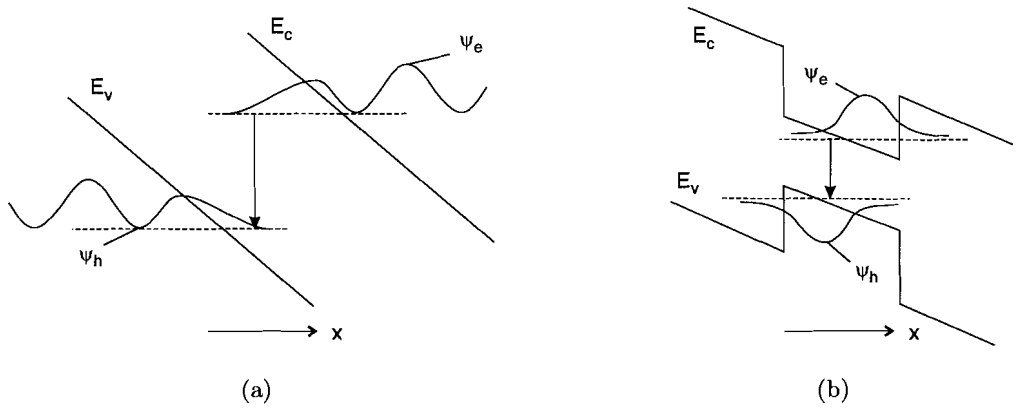


FIGURE 5.2: *Illustration of the Franz-Keldysh (a) and quantum confined Stark effect (b).*

THE FRANZ-KELDYSH EFFECT

Like the Stark effect, the Franz-Keldysh effect describes a change in absorption. When applying an electric field, the valence and conduction bands get tilted in real space. Consider figure 5.2(a). When no field is applied, an electron wave traveling in the x-direction will experience no change in the conduction band energy and can continue without perturbation. When a field is applied, the same electron wave will 'bump into' the higher energy level at a different location. This effectively changes the plane wave functions into Airy functions with a tail in the bandgap. This tail is a result of the finite probability of an electron penetrating into the field-induced barrier.

The same goes for holes in the valence bands. The finite overlap of the tails now indicates that a transition is possible below the energy of the bandgap. This quantum tunneling effect is also used in tunneling diodes. The larger the tilting of the bands, the higher the overlap of the Airy wavefunction tails. From Fermi's golden rule, the transition rate between the two states becomes larger. By calculating this overlap [30], the resulting change in absorption around the bandgap can be determined. More quantitative details will not be given here, as the resulting refractive index change is described by the Kerr effect.

THE QUANTUM CONFINED STARK EFFECT

Because of their coulombic interaction, a generated electron-hole pair can stay bound together. In doing so, they form a new quasi-particle called an exciton. As it requires less energy to generate an exciton than it takes to generate a free electron and hole, absorption slightly below the bandgap by an amount equal to the binding energy of the exciton is possible. The excitonic state with the highest binding energy is one that has a 1s-like state. In this state, the exciton has a binding energy of a few meV, and can only exist for a very short while (a few hundred fs at room temperature) in bulk material due to scattering and temperature effects. The energy uncertainty because of the Heisenberg uncertainty principle, is comparable to the binding energy. The exciton absorption will therefore not be resolvable in a bulk material like InP.

The exciton Bohr radius is rather large in bulk material (about 30 nm), and because of the cubic dependence of the Stark shift on the Bohr radius, a considerable Stark shift can be expected for moderate field strengths. Unfortunately, in applying this field, the exciton gets so quickly field-ionized that the whole concept of a bound state loses any useful meaning. In a quantum well however, the electron and hole are forced to stay together even under very high field strengths. This implies that very large Stark shifts can be obtained for excitons in a quantum well. This type of Stark shift is called the quantum confined Stark effect (QCSE).

In quantum confined structures like a quantum well, another effect plays a role when an electric field is applied. The field causes the electron and hole to get pulled to seperate sides of the quantum well due to their opposite charges. This results in a reduced overlap, causing the absorption strength at the original bandgap to reduce. Due to the penetration of the wavefunctions into the bandgap, transitions below the bandgap become possible so that the bandgap is effectively lowered. This effect is illustrated in figure 5.2(b). Much like a quantum-confined Franz-Keldysh effect, the wavefunctions now become Airy functions. The combined effects of the QCSE allow for large changes in absorption around the bandgap and similar changes in refractive index.

In an infinite potential well approximation using variational calculations [31] the magnitude of the energy shift is given by:

$$\Delta E = \frac{1}{8}\left(\frac{1}{3} - \frac{2}{\pi^2}\right)^2 \frac{me^2d^4}{\hbar^2}F^2 \tag{5.12}$$

where a positive ΔE is defined to be a shift resulting in a lower bandgap (redshift), m is the carrier effective mass (m_e, m_{hh} or m_{lh}), e is the electron charge, d the quantum well width and F the electric field strength. The fourth power dependence of the shift on the quantum well width has serious consequences. Considering heavy holes in InP with an effective mass of 0.6 m_0, $\Delta E \approx 0.13$ meV for a 3 nm quantum well, and $\Delta E \approx 17$ meV for a 10 nm quantum well, both at 100 kV/cm.

Because the effect is so small for a 3 nm quantum well like the one used in this work and because the optical overlap with the well is only in the order of 0.5%, we will neglect the QCSE and it will not be described in more detail here.

The Pockels effect

We have already seen from the Stark and Franz-Keldysh effect that applying an electric field changes the way bound electrons and free carriers behave in the crystal. The field does not only change the energy of the states but also the shape of their orbits. In doing so it will introduce an anisotropy in the originally isotropic crystal, changing the ϵ_r of the material from a scalar to a tensor. From Maxwell's equations it can be seen that this implies that now two solutions can be found for light traveling in a one direction, making the crystal birefringent. This effect is called the Pockels effect.

The amount of birefringence introduced by the Pockels effect is described by the electro-optic tensor, which has three non-zero elements of magnitude r_{41} for a noncentrosymmetric cubic crystal like InP. The elements all describe the change in refractive index in the directions perpendicular to that of the applied field. Therefore applying a field in the x-direction, as in our case, will have no influence on a polarization which has its electric field in the same

direction. This implies that TM polarized light does not get affected by the Pockels effect, whereas TE polarized under the right conditions does.

Because the POLIS material is highly absorbing for TE polarized light, the Pockels effect can unfortunately not be used for our application. It is worth noting that the difference in the Pockels effect for both polarizations is normally one of the problems encountered in designing polarization-insensitive electro-optical switches.

The Kerr effect

Similar to the Pockels effect, the Kerr effect can be described by an electro-optic tensor. The relevant coefficients of this tensor are s_{11} and s_{12}, where s_{11} describes the magnitude of Δn for a polarization with its electric field in the direction of the applied electric field and s_{12} describes the magnitude in the directions perpendicular to it. In the case of TM polarized light, the electric field components are mostly in the direction of the applied field, and therefore we have to consider s_{11}. The relation between the change in refractive index and this coefficient is:

$$\Delta n_{\text{kerr}} = \frac{1}{2} s_{11} n^3 F^2 \tag{5.13}$$

where n is the refractive index and F is the applied field in the x-direction.

The effect has a quadratic dependence on the applied electric field strength and because it's directly related to the quadratic Stark and Franz-Keldysh effects, it shows a strong dispersion near the band gap of the material. Some proposed models assume the Kerr effect arises due to either the Franz-Keldysh [32] or quadratic Stark [33] effect and try to match the theoretical models to experimental results. In this case we use a model purely fitted to experimental data, as the Kerr effect gives a phenomelogical description of the refractive index change. We use the model proposed in reference [34]. Here, s_{11} is obtained by fitting experimental curves:

$$s_{11} = C_{11} \frac{E^2}{n^4 (E_g^2 - E^2)^2} \, [\text{m}^2\text{V}^{-2}] \tag{5.14}$$

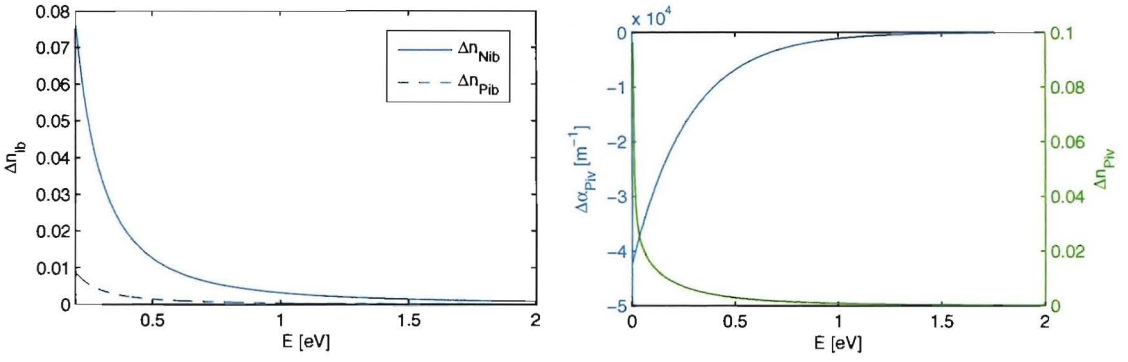
where $C_{11} = 3.2{\cdot}10^{-18} \, [\text{eV}^2\text{m}^2\text{V}^{-2}]$ is a fitting parameter determined in [34], n the refractive index, E_g [eV] is the bandgap of the material and E [eV] is the photon energy.

The plasma effect

This effect is directly related to the absorption mechanisms described in section 4.2. From reference [20] it is seen that there is an analytical description to account for the refractive index change due to intra-band absorption. As we will be considering the complete removal of carriers under a reverse bias, we can write:

$$\Delta n_{\text{ib}} = \frac{\chi e^2 h^2}{8\pi^2 \epsilon_0 m n E^2} \tag{5.15}$$

where χ is the doping level (P or N), e is the elementary electron charge, h is Planck's constant, ϵ_0 is the permittivity of a vacuum, m is the carrier effective mass (m_e or m_h), n is the refractive index of the material and E is the photon energy. The hole effective mass can be calculated from equation 4.5.



(a) (b)

FIGURE 5.3: *Change in refractive index as a function of photon energy due to intra-band absorption in InP (a) and inter-valence band absorption (b) versus photon energy, both at a doping level of 1e18* cm^{-3}.

In figure 5.3(a) the resulting change in refractive index for both p- and n-type doping in InP is given. Notice that due to the smaller effective mass of holes, the change in refractive index is much lower than for electrons.

This description does not however account for the inter-valence band absorption that also plays a significant role in p-doped materials. Only experimentally fitted data of the absorption change is available [21], that was described in section 4.2 by equation 4.6. We obtain the related Δn by applying the Kramers-Kronig transform:

$$\Delta n_{P\mathrm{iv}} = \mathcal{K}\{\Delta\alpha_{P\mathrm{iv}}\} = \mathcal{K}\{-4.252 \cdot 10^{-20} e^{-2.28 \cdot 10^{19} E} P\} \tag{5.16}$$

with the photon energy E and P the hole concentration.

In figure 5.3(b) the change in absorption and refractive index as calculated with equation 5.16 for P = 1e18 cm^{-3} is given. From this figure it is clear that the change in refractive index is largest at energies with a large change in absorption. The Δn for holes due to intra-band absorption is much smaller than due to the inter-valence band absorption. This was to be expected from the magnitude of these absorption coefficients. As the two effects are independent, their contribution to Δn can be summed:

$$\Delta n_{P\mathrm{plasma}} = \Delta n_{P\mathrm{ib}} + \Delta n_{P\mathrm{iv}} \tag{5.17}$$

$$\Delta n_{N\mathrm{plasma}} = \Delta n_{N\mathrm{ib}} \tag{5.18}$$

THE BANDGAP SHRINKAGE EFFECT

This effect is also known as the bandgap renormalization or bandgap narrowing effect, and is caused by the Coulomb screening effect of free carriers. Due to carrier injection or, in our case, doping, there will be carriers that are free to move around in the valence or conduction band. Assume now that a photon is absorbed and a newly generated free electron enters this 'pool' of free electrons. Because of the Coulomb interaction between the electron and surrounding electrons, the existing electrons will redistribute themselves to minimize the potential energy

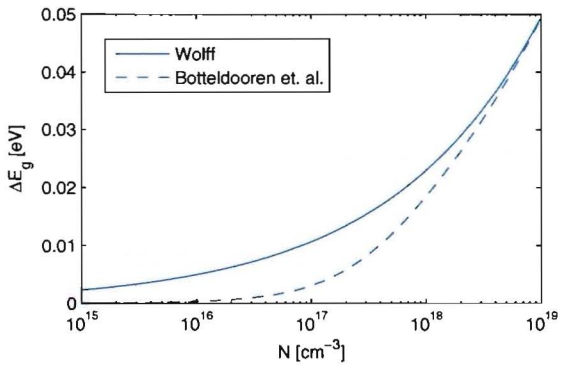


FIGURE 5.4: *Models describing the magnitude of the bandgap shrinkage effect in n-doped InP. Taken from reference [25] (solid) and reference [27] (dashed).*

between them. This redistribution or screening of neighboring electrons effectively reduces the energy of the new electron and shifts the conduction band edge down. A transition with a photon energy slightly below the bandgap energy is now possible, effectively lowering the bandgap. The same reasoning goes for holes in the conduction band.

There is an expression available for the magnitude of the bandgap shrinkage, derived in reference [25]:

$$\Delta E_g = \frac{e^2}{2\pi\epsilon}\sqrt[3]{\frac{3\chi}{\pi}} \tag{5.19}$$

where a positive ΔE_g is defined to be a shift to a lower bandgap (redshift), χ is the carrier density (holes or electrons). This expression is only assumed valid for carrier densities where the inter-carrier spacing is equal or less than the Bohr radius of these carriers [26]. For electrons in InP, it is in the order of 1e18 cm^{-3}.

In reference [27] it is pointed out that the $\sqrt[3]{\chi}$ behavior is a $T = 0$ K feature, and a correction is made for $T = 300$ K. They fit their analytically obtained bandgap shrinkage for InGaAsP compositions according to:

$$\Delta E_g = \frac{\sqrt{me}}{\epsilon_r}\frac{1.04{\cdot}10^3(\frac{\chi}{m^{3/2}})^{-0.19} + 2.8{\cdot}10^{-7}(\frac{\chi}{m^{3/2}})^{\frac{1}{3}}}{1 + \frac{2.4{\cdot}10^{19}m^{3/2}}{\chi}} \tag{5.20}$$

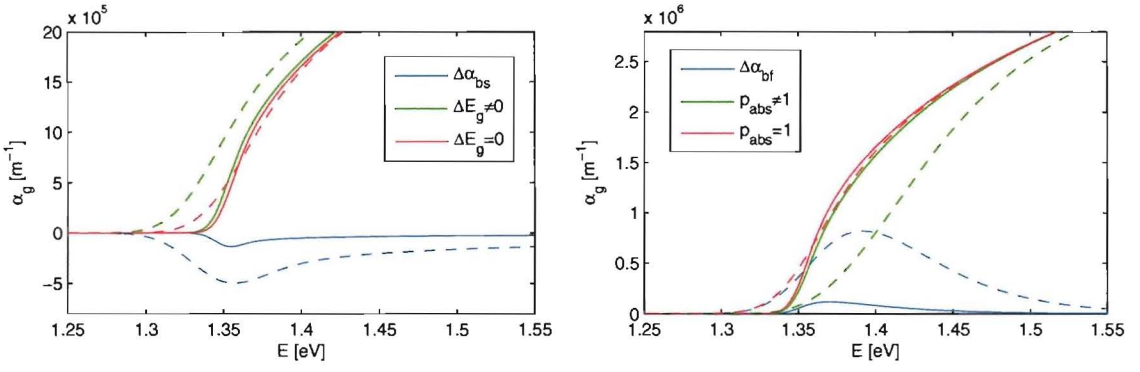
where ϵ_r [ϵ_0] is the relative permittivity of the material and for p-type doping χ [cm^{-3}] and m [m_0] should be replaced by P and m_h and similarly by N and m_e for n-type doping.

In figure 5.4, the ΔE_g from equations 5.19 and 5.20 for n-InP are shown. Equation 5.20 obviously approaches equation 5.19 for high doping levels. We will use 5.20 for our model, as it has proven to give more accurate results.

The calculated ΔE_g can easily be incorporated in equation 5.8 by replacing E_g by $E_g - \Delta E_g$. The effective change in absorption is obviously the final absorption curve minus the initial one, and as we will be considering the removal of carriers under a reverse bias we can write the following resulting change in refractive index:

$$\Delta n_{\mathrm{bs}}(E) = \mathcal{K}\{\Delta\alpha_{\mathrm{bs}}\} = \mathcal{K}\{\alpha_g - \alpha_g|_{E_g-\Delta E_g}\} \tag{5.21}$$

where $|_{E_g-\Delta E_g}$ indicates that E_g in equation 5.8 should be replaced by $E_g - \Delta E_g$.



(a) (b)

FIGURE 5.5: *Absorption change under the removal of carriers in n-InP due to bandgap shrinkage (a), and bandfilling (b). Solid: 1e17 cm^{-3}, dashed: 1e18 cm^{-3}.*

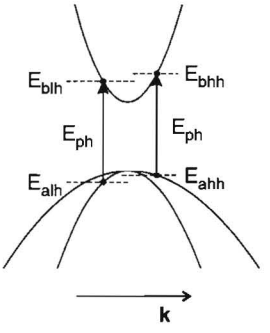


FIGURE 5.6: *Transition probabilities and corresponding energy levels.*

In figure 5.5(a), the curves of α_g ($\Delta E_g = 0$), $\alpha_g|_{E_g - \Delta E_g}$ ($\Delta E_g \neq 0$) and the resulting $\Delta\alpha_{\text{bs}}$ for a N of 1e17 cm^{-3} (solid lines) and 1e18 cm^{-3} (dashed lines) are given for InP. The resulting Δn will be considered together with the bandfilling effect described next.

THE BANDFILLING EFFECT

In both bulk and quantum well material, the amount of allowable energy states in valence and conduction band is limited. If there are many carriers present, they will fill up the lowest of these available energy states in a band, leaving no vacancies for new carriers at these locations. This effectively decreases the absorption strength of the material material above the bandgap.

Consider figure 5.6. Assume that there is a photon with sufficient energy E_{ph} available to get absorbed by exciting an electron. This electron is then moved from the valence band to the conduction band. The probability that this happens p_{abs} is the probability that there is an electron available at the initial energy level E_a in the valence band, times the probability that there is already an electron occupying the final energy level E_b in the conduction band. The

resulting probability can be approximated by the Fermi-Dirac distribution function, given in appendix F.

In an undoped semiconductor without injection, virtually all electrons are in the valence band and none in the conduction band, so that $p_{\text{abs}} = 1$. With p-type doping there are more holes present in the valence band, and in the case of n-type doping there are more electrons in the conduction band and thus p_{abs} will always drop below unity. p_{abs} determines by how much the absorption at a certain energy level is decreased. The resulting change in refractive index under the removal of carriers can be written as:

$$\Delta n_{\text{bf}} = \mathcal{K}\{\Delta\alpha_{\text{bf}}\} = \mathcal{K}\{\alpha_g - \alpha_g \cdot p_{\text{abs}}\} \tag{5.22}$$

where again α_g is the absorption around the bandgap according to equation 5.8.

As we consider two valence bands (heavy and light hole), p_{abs} will be different for both. To incorporate this, the absorption parameter C described in section 5.2 is split into two parts. One involving light holes C_{lh}, and another for heavy holes C_{hh}:

$$\Delta n_{\text{bf}} = \mathcal{K}\{\Delta\alpha_{\text{bf}}\} = \mathcal{K}\{\alpha_g - [\alpha_g|_{C_{lh}} \cdot p_{\text{abs}lh} + \alpha_g|_{C_{hh}} \cdot p_{\text{abs}hh}]\} \tag{5.23}$$

where $|_{C_{lh}}$ indicates that C in equation 5.8 should be replaced by C_{lh}.

The values of the various p_{abs} are derived in appendix F, according to the model in reference [22]. C_{lh} and C_{hh} are determined by considering that the relevant parameters in the absorption process are the reduced effective masses of the electron hole pairs, given by:

$$m_{elh} = \frac{m_e m_{lh}}{m_e + m_{lh}} \tag{5.24}$$

$$m_{ehh} = \frac{m_e m_{hh}}{m_e + m_{hh}} \tag{5.25}$$

Because the concentrations of heavy and light holes are directly related to their effective masses, we thus have:

$$C_{lh} = \frac{m_{elh}^{3/2}}{m_{elh}^{3/2} + m_{ehh}^{3/2}} C \tag{5.26}$$

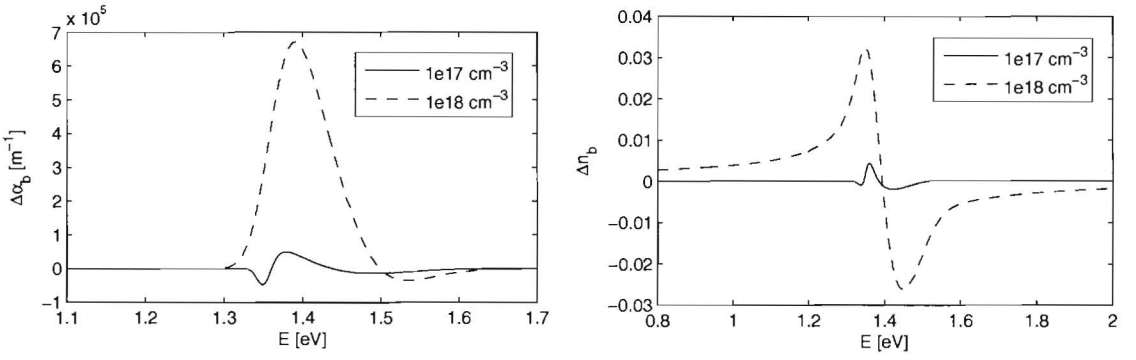
$$C_{hh} = \frac{m_{ehh}^{3/2}}{m_{elh}^{3/2} + m_{ehh}^{3/2}} C \tag{5.27}$$

The resulting changes in the absorption profile are given in figure 5.5(b) for n-type doping in InP. The bandgap is not corrected by the ΔE_g of the bandgap shrinkage effect. The effect is much less for p-type doping, due to the fact that the density of states in the valence bands is much higher than in the conduction band (i.e. the conduction band gets filled 'quicker' than the valence bands).

The bandgap shrinkage and bandfilling effect combined

We can now combine the bandgap shrinkage and bandfilling in the following way:

$$\Delta n_{\text{b}} = \mathcal{K}\{\Delta\alpha_{\text{b}}\} = \mathcal{K}\{\alpha_g - [\alpha_g|_{E_g - \Delta E_g, C_{lh}} \cdot p_{\text{abs}lh} + \alpha_g|_{E_g - \Delta E_g, C_{hh}} \cdot p_{\text{abs}hh}]\} \tag{5.28}$$



(a) (b)

FIGURE 5.7: *Change in absorption due to the combined bandgap shrinking and bandfilling effect (a) and the related change in refractive index (b) in InP.*

where the subscript b stands for the combined bandgap shrinkage and bandfilling effect.

Another term should be included in this description, as from experimental data [15] it follows that the absorption doesn't change at energies more than 0.2 eV above the bandgap energy, even for very high doping levels. To incorporate this given fact, we add a quickly decaying exponential at 0.1 eV above the bandgap, ensuring that $\Delta\alpha \approx 0$ at $E > E_g + 0.2$ [eV]:

$$\Delta n_b = \mathcal{K}\{\Delta\alpha_b\} \qquad E \leq E_g + 0.1 \tag{5.29}$$

$$\Delta n_b = \mathcal{K}\{\Delta\alpha_b e^{\frac{[E_g-(E+0.1)]^2}{0.1^2}}\} \qquad E > E_g + 0.1 \tag{5.30}$$

Here all energies in eV, and Δn_b and $\Delta\alpha_b$ can be replaced by their p- or n-type variants. This adaptation is verified by comparing with experimental data from reference [15].

The bandgap shrinkage and bandfilling effect are in a way competing with eachother, especially for lower doping levels. This is clearly illustrated in figures 5.7(a) and 5.7(b).

5.4. ELECTRIC FIELD UNDER A REVERSE BIAS

To be able to calculate the amount of refractive index change due to the above mechanisms, we have to determine which part of the semiconductor structure is affected by an existing electric field. I.e. $F \neq 0$ and carriers swept out of the depletion region. Considering a p-n junction at $x = 0$, we know from Poisson's equation in one dimension that:

$$F(x) = \int \frac{\rho(x)}{\epsilon(x)} dx \tag{5.31}$$

where $F(x)$ is the electric field strength distribution, $\rho(x)$ is the volume charge density distribution and $\epsilon(x)$ is the permittivity of the material at position x. We assume abrupt junctions with uniform doping levels and n-type background doping for intrinsic materials, so that we obtain:

$$\rho(x) = -eP(x) \qquad \text{p} - \text{type} \tag{5.32}$$

$$\rho(x) = eN(x) \qquad \text{n} - \text{type} \tag{5.33}$$

where P and N are the doping levels. The built-in potential of the p-n junction is then given as:

$$V_{bi} = -\int_{-x_p}^{x_n} F(x)dx \tag{5.34}$$

where x_p and x_n are the depletion widths in the p- and n-doped regions. $F(x)$ will be zero outside this space charge region.

In the above formulas the fact that the electric field can stretch out into different materials, is incorporated by making ϵ, P and N position dependent. To determine x_n and x_p, we have to obtain V_{bi} by other means. The magnitude of the built-in potential is determined by the difference of the Fermi-levels in both the p- and n-type material, thus:

$$V_{bi} = E_{NF} - E_{PF} \tag{5.35}$$

according to the definition of the levels given in appendix F.

A solution for $F(x)$ can be found by determining V_{bi} using equation 5.35 and iteratively increasing x_p and x_n. The corresponding V_{bi} is then calculated according to equation 5.34 until both match. This numerical approach allows us to find a solution for various layer stacks. In applying a reverse bias V_b the same procedure is followed, but now by finding a solution for $V_{bi} + V_b$.

A heterostructure junction introduces a small jump in $F(x)$ at the p-n junction due to the band offset. For InGaAsP heterojunctions of different compositions this jump is very small and rather hard to calculate, so it will be neglected here. The model was compared with an existing simulation program (SimWindows) and show an excellent agreement in the description of the electric field distribution.

5.5. Phase shifter

Now that we know the distribution of the electric field, we can calculate the $\Delta n(x)$ at every position in the structure by calculating the appropriate refractive index change mechanisms. For the Kerr effect, the change of the electric field has to be considered at a given position, whereas for carrier effects, the change in depletion determines whether it will contribute. We can add up the separate effects as they are independent, giving:

$$\Delta n = \Delta n_{\text{kerr}} + \Delta n_{\text{plasma}} + \Delta n_{\text{b}} \tag{5.36}$$

The effective change in refractive index at a voltage V for light traveling through the structure is determined by the overlap of its optical intensity field $I(x)$ with the local $\Delta n(x)$ at that position:

$$\Delta n_{\text{eff}}(V) = \frac{\int_{-\infty}^{\infty} \Delta n(x) I(x) dx}{\int_{-\infty}^{\infty} I(x) dx} \tag{5.37}$$

$I(x)$ can be calculated using the method described in appendix C, neglecting the influence of the small Δn on the distribution of the optical field and assuming all optical power is contained within I_x (i.e. a pure TM polarization, see section 4.2). The optical intensity is normalized to unity, so that we can simplify the above expression to:

$$\Delta n_{\text{eff}}(V) = \int_{-\infty}^{\infty} \Delta n(x) I(x) dx \tag{5.38}$$

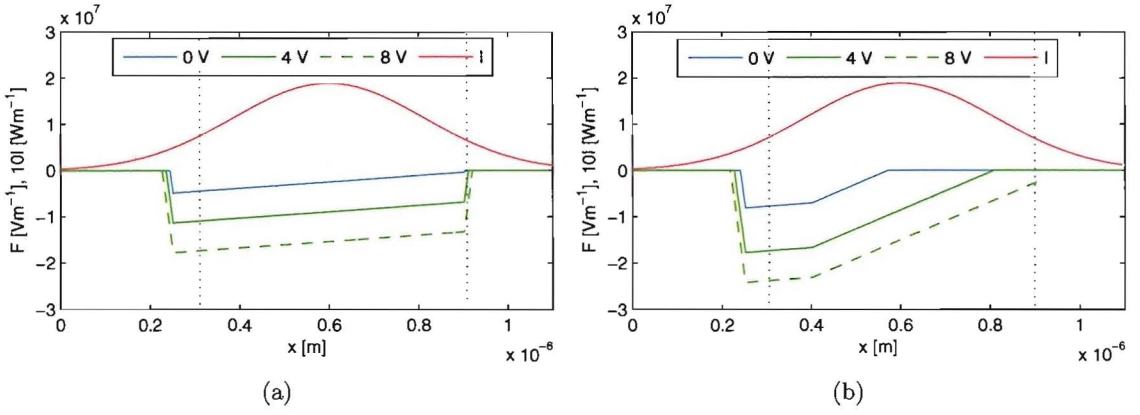


FIGURE 5.8: *Optical and electric field distributions in the phase shifting section for AX06-029 (a) and AX06-030 (b), for various reverse bias voltages. Dotted lines indicate the position of the waveguide material.*

In these calculations the exact distribution of the fields in the y-direction is assumed to be unimportant; i.e. in the y-direction the light experiences the same electric field strength everywhere. This is a valid approximation under normal circumstances, where the electric field exists over a depletion region under a ridge and virtually all of the light propagates under this ridge [36].

The Δn_{eff} that we can use for switching is given by:

$$\Delta n_{\text{sw}} = \Delta n_{\text{eff}}(V_b + V_{bi}) - \Delta n_{\text{eff}}(V_{bi}) \tag{5.39}$$

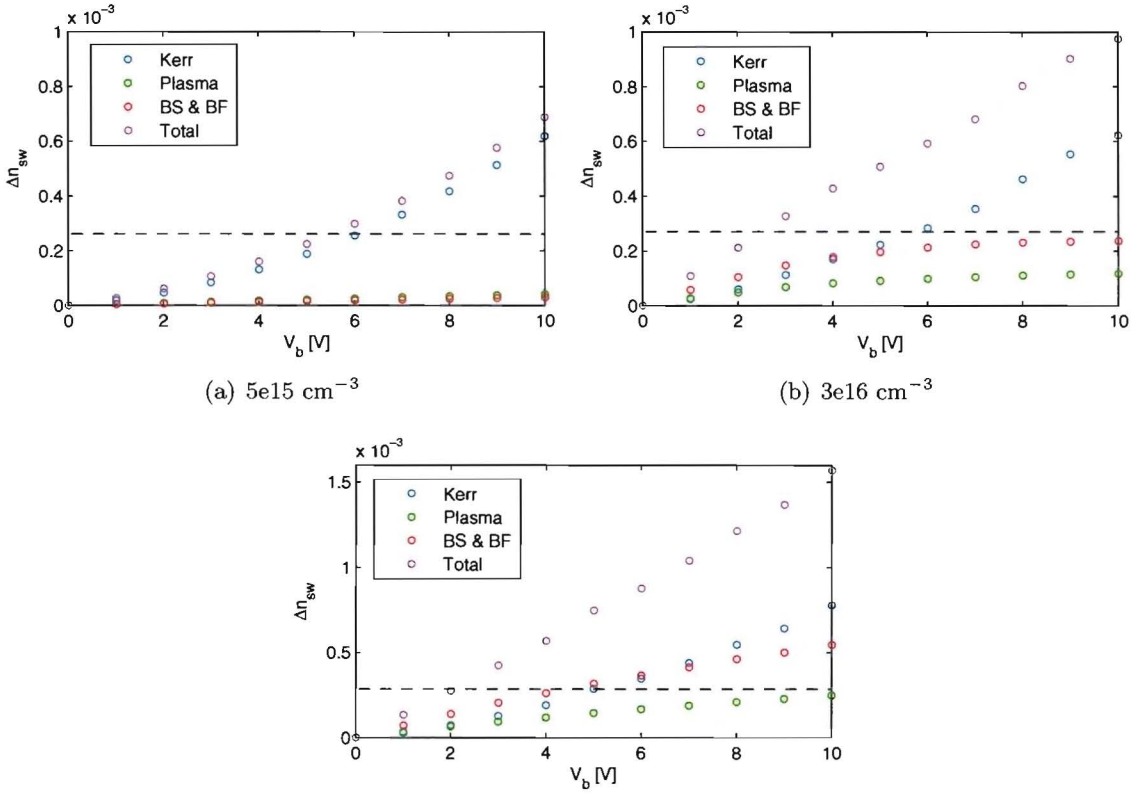
This implies that Δn under a reverse bias should be corrected by the value of Δn due to the built-in potential.

An illustration of the optical field and electric field distribution is given in figure 5.8(a) and 5.8(b), for *AX06-029* and *AX06-030* respectively. The optical field strength is exaggerated by a factor 10 for clarity. Note that in the case of *AX06-029*, the waveguide region is already depleted without applying a reverse bias.

When the light propagates through two arms (labeled a and b) of an MZI, the effective path length difference the light experiences is $(n_a - n_b)L = \Delta n_{\text{sw}} L$, where n_a and n_b are the effective refractive indices of the two paths and L is the length. We obtain switching with a π phase difference, which equals an optical path length difference of $\frac{\lambda_0}{2}$. The required phase shifting length L_π is therefore:

$$L_\pi = \frac{\lambda_0}{2\Delta n_{\text{sw}}} \tag{5.40}$$

In figures 5.9(a) to 5.9(c), values of Δn_{sw} are given for increasing reverse bias voltages. The dashed line indicates the required Δn for a π phase shift using a 3 mm phase shifting section. It is clear that the slightly higher doping of *AX06-030* is superior over that of *AX06-029*, because in the latter the waveguide region is already depleted without a reverse bias. This minimizes the Δn effects due to carriers. The figures predict that the required V_π is 5.4 V for *AX06-029* and 2.6 V for *AX06-030*, which is a significant improvement. Increasing the doping in the bottom waveguide layer (layer 5 in table 4.2) even further to 1e17 cm^{-3}, does



(a) 5e15 cm^{-3}

(b) 3e16 cm^{-3}

(c) 1e17 cm^{-3}

FIGURE 5.9: *Simulated effective refractive index change at different reverse bias voltages for three different doping levels of layer 5 in table 4.2. (a) corresponds to AX06-029, (b) corresponds to AX06-030.*

give a slight additional decrease of V_π to 2.0 V, but will also increase the losses significantly. Therefore the moderate doping of 3e16 cm^{-3} was chosen for *AX06-030*.

The importance of keeping the V_π low comes from absorption effects. As mentioned before, refractive index changes are always paired with absorption changes. When applying high bias voltages, these absorption effects will become significant. As a general rule, the switching voltage should stay under 5 V.

5.6. Conclusions

In this chapter, we have presented the electro-optical effects that we can apply in achieving an effective phase shifter for our EOS. These effects turn out to be the Kerr, plasma, bandgap shrinkage and bandfilling effect. We combined the last two to facilitate calculations. We found that the expected required reverse switching voltages are 5.4 V and 2.6 V for a switch with a 3 mm phase shifter realized on *AX06-029* and *AX06-030* respectively. The main reason for this difference is that for *AX06-029* the waveguide is already fully depleted under zero bias, so that carrier effects play an insignificant role.

The model used differs from those presented in literature in the following ways:

- The inter-valence band absorption is included in calculating the plasma effect. In other models, only the intra-band absorption is considered.
- A proper description of the absorption around the bandgap is used, including the Urbach tail.
- Changes of the absorption around the bandgap due to carrier effects are limited to 0.2 eV above the bulk bandgap, as confirmed by experiments [15].
- The bandgap shrinkage is not ignored completely [36, 23], ignored below a critical doping level [22], determined a posteriori [34] or determined by a high-density approximation [25] . Instead we use a model that corrects for leaving the high-density limit [27], which is appropriate for operation at 300 K.

6

Design, realization and characterization of the POLIS switch

6.1. Introduction

In the previous chapter we have determined the magnitude of the electro-optical effects and we have seen that a sufficient effect can be achieved using a 3 mm phase shifting section. We now turn to the design, realization and characterization of complete electro-optical switches. As an existing design has been used, we first discuss the existing layout and describe how it was realized. We then briefly address the different parts of the design and their design implications. Finally the behavior of the realized switches is evaluated, comparing it with the model.

6.2. Mask layout

Unfortunately, a new set of lithographic masks to realize an optimal design of the switch could not be made due to time constraints. We therefore had to turn to an existing switch design for which the lithographic masks were available. There was only one design available suitable for our wafers which have a semi-insulating substrate and need top n-contacts. This design has the switches under a 45° angle with the cleavage planes, connected with long waveguides. Furthermore, most of these waveguides end with special structures (tapers and spot-size converters) at the cleavage planes, which needs special fiber arrays for light coupling. And finally only two sets of switches feature top n-contacts.

The first two problems were overcome by tilting the masks under a 45° angle during all lithography steps, and precise cleaving of the processed chip. This is shown in figure 6.1(a), where the portions removed by cleaving are indicated by the arched regions. The green areas indicate waveguides, the purple is metal. For clarity, only a few of the mask layouts are displayed and non-relevant structures have been removed. This leaves us with four useable parts, effectively leaving 7 switches and waveguides with and without metal. These last were

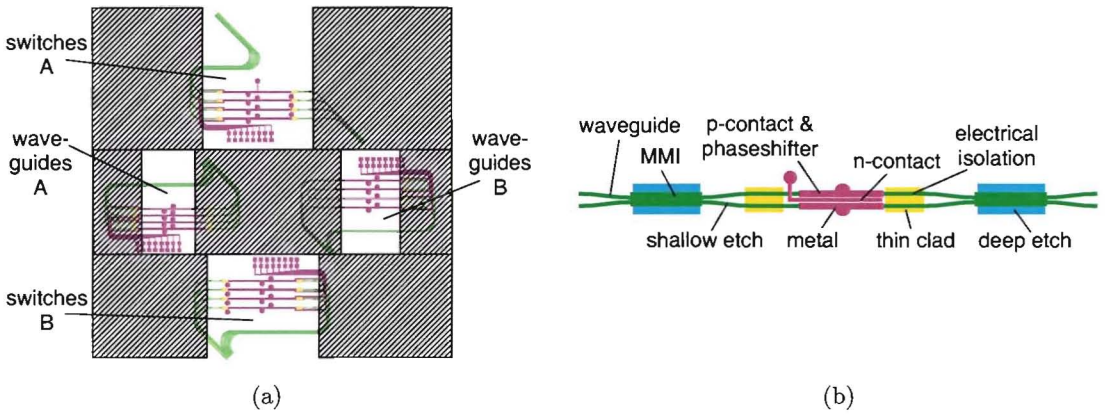


FIGURE 6.1: *Mask layout of the switches where the arched regions indicate the regions removed by cleaving (a) and layout of one switch (b).*

used for loss characterization.

6.3. REALIZATION

As we will be using an existing switch layout and corresponding process scheme, we first briefly discuss these. In the next section the design considerations will be given.

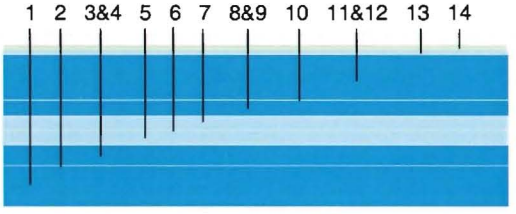
In figure 6.1(b) the layout of one switch is given. For clarity, the phase shifting section is depicted to be shorter than it is in reality. The colors indicate regions that are processed differently: shallowly etched, deeply etched, top cladding removed and metal deposited. The layout is basically that of a standard MZI switch as given in figure 1.2(a).

The most important processing steps to realize the switch are given in figure 6.2 (colors do not correspond to those in figure 6.1(b)). All processing was done using established methods in the group and will be described here briefly.

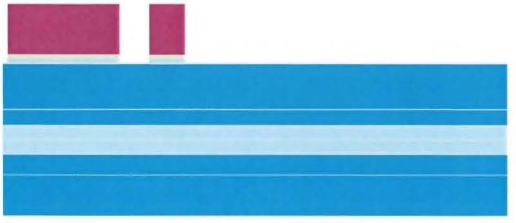
After removing the InP protection layer with a selective wet etch (figure 6.2(a)), a positive resist is spun, followed by a short prebake. The first mask exposure defines the regions where the contact layers stay intact and is followed by development of the resist. A selective etch removes the InGaAs and InGaAsP contact and diffusion stop layers (figure 6.2(b)). The resist is then removed by spraying with acetone and isopropanol. Remaining resist is removed in the stripper.

SiN_x is deposited by PECVD, and is made hydrophilic by oxidizing in the stripper. Using a primer, the surface is made hydrophobic so that photoresist can be spun on the silicon nitride surface. After lithography and reactive ion etching of the SiN_x, the photoresist is again removed. A new layer of resist is then spun in the same manner, followed by exposure of a lift-off profile defining the deep etched waveguides. Titanium is evaporated by electron beam evaporation and the lift off is performed using acetone (figure 6.2(c)).

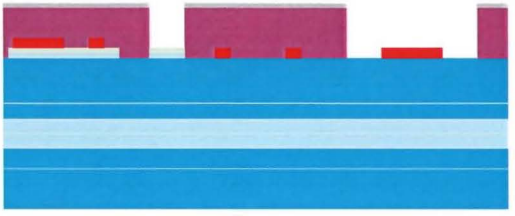
During a RIE etch, an amount of material is removed so that the waveguide layer is exposed in these regions. The titanium and resist are removed afterwards (figure 6.2(d)). Another RIE etch removes another similar amount of material, defining shallow and deep waveguides



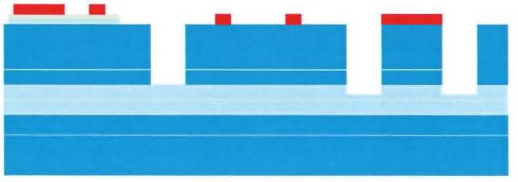
(a) Protection layer removed.



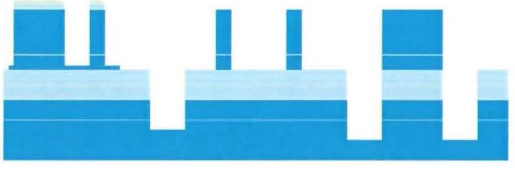
(b) Resist & wet etch for contacts.



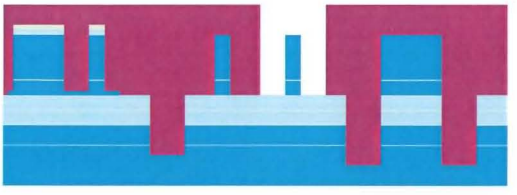
(c) SiN_x & resist for shallow and deep waveguides.



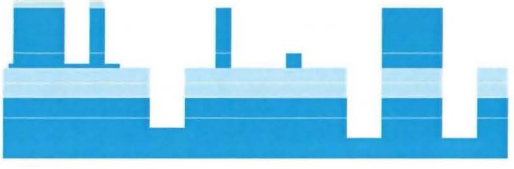
(d) RIE etch for deep waveguides.



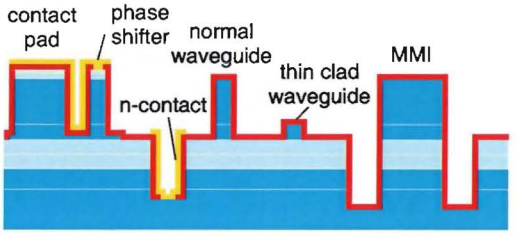
(e) RIE etch for shallow and deep waveguides.



(f) Resist for thin clad waveguides.

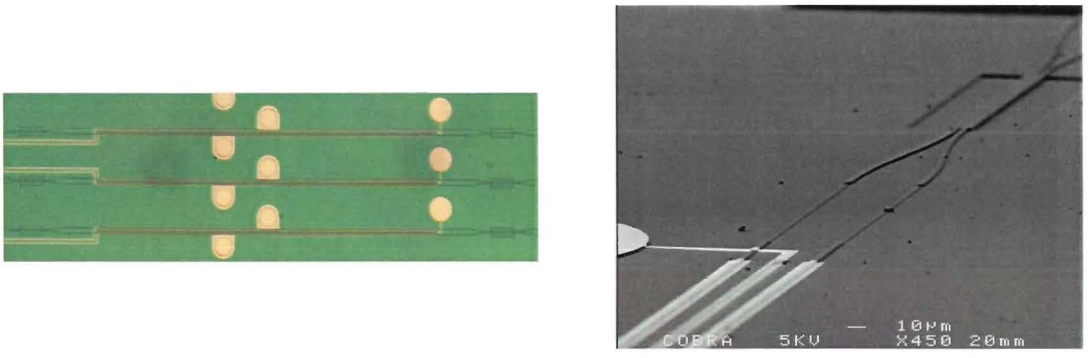


(g) Wet etch for thin clad waveguides.



(h) SiN_x & metal for passivation and contacts.

FIGURE 6.2: *Process flow of the switch realization. Numbers indicate the layers as given in table 4.2. Blue: InP, lightblue: Q(1.25), lightgray: InGaAs, purple: resist, red: SiN_x, darkgray: Ti, yellow: Ti/Au.*



(a) (b)

FIGURE 6.3: *Microscope (a) and SEM (b) image of the realized switches.*

and n-contacts. A few cycles of plasma stripping and oxide removal with phosphoric acid are used to remove the roughness caused by the RIE etch (figure 6.2(e)).

The subsequent lithography defines the regions where the cladding is removed for the electrical isolation of the phase shifters (figure 6.2(f)). A selective wet etch removes the top cladding (figure 6.2(g)).

A layer of SiN_x is deposited for passivation and p-contact openings are made by lithography and a SiN_x RIE etch (figure 6.2(h)). After removal of the resist, titanium and gold are evaporated. Another lithography protects the regions where the metal has to stay and the remaining metal is removed by wet etching (figure 6.2(h)). The last step is a short anneal to make ohmic contacts.

Processing did not give any problems, except for a small mistake during the second RIE etch (figure 6.2(e)). The RIE etching speed in the quaternary material is slower than that in InP, which was overlooked for *AX06-029*. The n-contact layer was not reached and the lithography and etch step had to be repeated, introducing a small mismatch and non-ideal profile for the MMI's. Furthermore, the surface of the *AX06-029* sample was scratched during the realization, destroying a whole set of switches. Eventually only two working switches were realized on this wafer. For *AX06-030* all switches were fabricated successfully and were found to work. In figures 6.3(a) and 6.3(b) a microscope and SEM image of the realized switches is given.

6.4. SWITCH DESIGN

WAVEGUIDES

All waveguides are shallow (green regions in figure 6.1(b)), meaning that the material surrounding the waveguide isn't etched through the waveguiding material. The guiding therefore is purely done on the basis of effective index differences. A shallowly etched waveguide has the advantage that losses are usually lower than for deeply etched waveguides. The light does not experience sidewall roughness as much as in the deep etched case. A disadvantage is that the lateral confinement is less, making it more sensitive to crosstalk when two or more

waveguides are closely spaced.

Bends are limited to a radius of about 500 μm, where for deep etched waveguides these can be a factor of 3 smaller. In such a bent waveguide, the mode propagates slightly more towards the outside of the bend. To facilitate transitions of straight to bent waveguides (and vice versa) and bent to oppositely bent waveguides, an offset of 0.35 μm and 0.7 μm respectively is introduced between these waveguides.

The guides have a designed width of 3.3 μm, but the smoothing procedures to remove sidewall roughness will slightly decrease the actual width. They will support the fundamental, first and second order modes. The reason for a rather wide waveguide is to relax the alignment precision demands in making a contact opening on top of the waveguide for the phase shifting section during a subsequent lithography step. A careful injection of light into the waveguides can avoid excitation of the higher order modes.

MMI's

The employed 2x2 MMI's are of the restricted interference type [3], so that only certain modes in the MMI are excited. This reduces the required length of the MMI by a factor of 3 compared to a general 2x2 MMI. This also implies that the center of the in- and output waveguides have to be spaced at a distance of $\pm\frac{1}{6}W_{\mathrm{MMI}}$ from the center of the MMI, where W_{MMI} is the MMI width. The MMI is 12 μm by 200 μm, so that the spacing between the waveguides is only 4 μm. Some crosstalk at the in- and output is then possible. To limit this effect, the waveguides are separated directly by bends.

To increase the number of supported modes which is needed for low crosstalk operation of paired interference MMI's, they are deeply etched (the blue region in figure 6.1(b)). Because the design of the MMI is based on fundamental-mode excitation, higher order modes entering the MMI from the input waveguides will result in higher loss and crosstalk.

Phase shifter

The phase shifting sections have a length of 3 mm, to insure a sufficient refractive index change before absorption effects that occur at high reverse biases start to play a role. A small ridge of 1.7 μm in width is opened in the SiN_x on the waveguides and the p-contact is made here. The n-contact for the phase shifter is made by etching to the heavily doped n-contact layer. Large contact pads allow for an easy connection using probes.

Thin clad waveguides

In applying a reverse bias on one phase shifter, the other phase shifter branch is in principle also electrically connected through the thick highly doped top cladding. This path runs from the p-contact through the p-doped cladding, via the MMI back to the other branch. Because of this, part of the voltage drop can occur at this other branch giving less differential refractive index change and effectively reducing the efficiency of the switch. To overcome this problem, the top cladding above the first etch stop layer is removed in a small region (yellow in figure 6.1(b)). This breaks the circuit and assures that the voltage drop will occur only in one phase

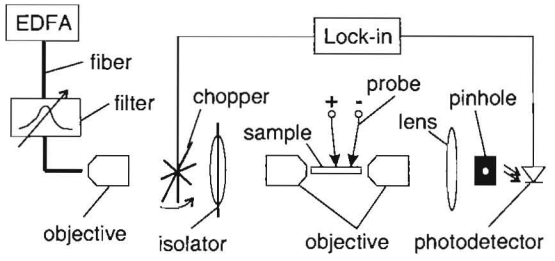


FIGURE 6.4: *Measurement setup.*

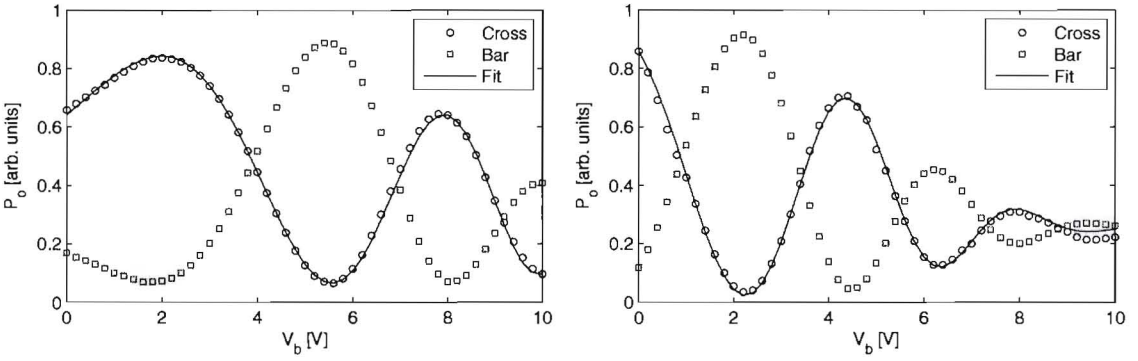


FIGURE 6.5: *Switching behavior of a switch realized on AX06-029 (a) and AX06-030 (b) for increasing applied reverse bias voltages.*

shifting branch.

6.5. CHARACTERIZATION

SWITCHING EFFICIENCY

In figure 6.4, a schematic representation of the measurement setup is given. Light is generated by a broad band source (EDFA) and bandpass filtered. It exits the fiber to free space and the beam is collimated by a microscope objective. It then passes through a chopper connected to a lock-in amplifier, and an isolator to select the TM polarization. The light is then focussed onto the sample, so that it couples into a waveguide. The sample is biased by a controlled voltage source using probes. The light exiting from one of the two output ports is again collimated by a microscope objective and focused onto a photodetector by a lens. Background light is eliminated by a pinhole. The photodetector is also connected to the lock-in amplifier, from which we read the output light intensity.

In figures 6.5(a) and 6.5(b) the output power in the cross and bar state for a switch on *AX06-029* and *AX06-030* as a function of applied reverse bias on one phase shifter is shown.

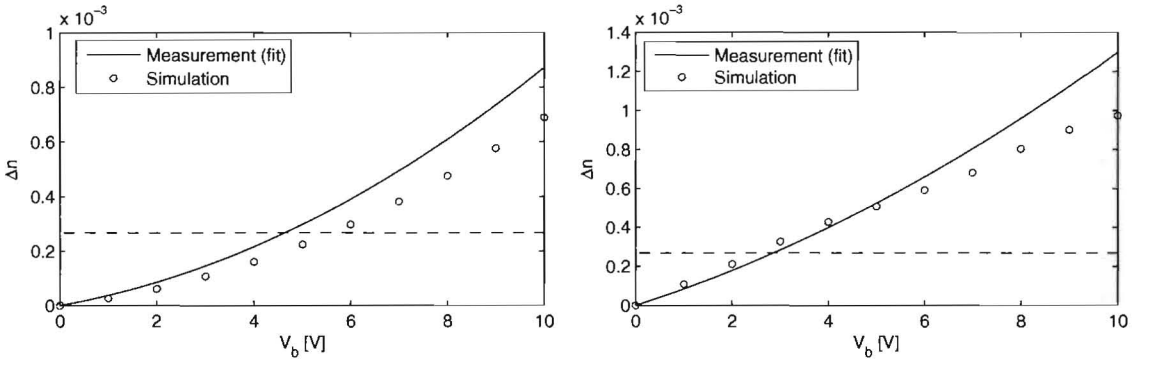


FIGURE 6.6: Δn_{eff} *as derived from the fitting procedure and the corresponding simulation values for a switch on AX06-029 (a) and AX06-030 (b).*

The curves were obtained by transmission measurements, using an isolator to set the input polarization to TM. The switching characteristics of the most efficient devices found are shown here. A fit to the cross-state output according to the model described in appendix 2D is also given in these figures.

The fact that the curves do not have their maximum for $V_b = 0$ indicates that there is a slight fabrication induced phase offset ($\delta\varphi$, e.g. due to variation in the width of the phase shifting sections). This can be resolved by applying a fixed bias on one of the branches, while using the other for switching, but this wasn't done here. The sine-like behavior of the curve follows directly from equation 48 in appendix 2D, for increasing $\delta\varphi$.

The effective change in refractive index and the amount of phase difference due to a bias are directly related via:

$$\Delta n_{\text{sw}}(V) = \frac{\lambda_0}{2\pi L}\delta\varphi(V) \tag{6.1}$$

where $\Delta n_{\text{sw}}(V)$ is the bias-dependent effective refractive index change, $\delta\varphi(V)$ the bias-dependent phase shift, λ_0 is the wavelength of the light in a vacuum and L is the phase shifter length. Because we know the refractive index change mechanisms either have a linear or quadratic nature, we can obtain a good fit using a second order description of $\delta\varphi(V)$. The resulting fit is given in figures 6.5(a) and 6.5(b).

From the fit parameters and equation 6.1, Δn_{sw} can be derived. It is given in figures 6.6(a) and 6.6(b) together with the simulation results described in chapter 5. The required Δn_{sw} is indicated by the dashed line. The resulting V_π is 4.6 V and 2.8 V for a switch with a 3 mm phase shifter on *AX06-029* and *AX06-030* respectively, corresponding to a phase shifting efficiency of 13.0 °/Vmm and 21.4 °/Vmm. The predicted values from the simulation were 5.4 V and 2.6 V. Although not as large as the simulations predicted, the benefit from the slight intentional doping of *AX06-030* is obvious.

In both cases, the simulations underestimate the refractive index change effect for higher reverse biases. The most likely reason for this is that the strength of the Kerr effect is underestimated in the simulations and we can correct for this by increasing the coefficient in equation 5.14 by a factor of 1.3.

Taking another look at figures 6.5(a) and 6.5(b), we see that absorption effects start playing a role at lower voltages for *AX06-030* than for *AX06-029*. The reason for this is that the Franz-Keldysh effect greatly increases the absorption below the bandgap at high field strengths. Comparing figures 5.8(a) (*AX06-029*) and 5.8(b) (*AX06-030*) we see that the peak field strength is higher for the latter at the same reverse bias. The peak field strength for *AX06-030* at 4 V reverse bias is comparable to that of *AX06-029* at 8 V bias and are in the order of 200 kV/cm. In [18] the Franz-Keldysh effect can be seen to give band shifts of more than 140 meV at these field strengths which increases the absorption at the operating wavelength. It is this effect that diminishes the amplitude of the sinusoidal curves in figures 6.5(a) and 6.5(b) at an increasing reverse bias.

Crosstalk

When corrected for the fabrication induced phase offset, the crosstalk (or on-off ratio) is defined as the ratio of the amount of light that exits through the unwanted output port (the bar port for $V_b = 0$ and cross port for $V_b = V_\pi$) over the amount of light that exits through the wanted output port (cross port for $V_b = 0$ and bar port for $V_b = V_\pi$). For *AX06-029* the best obtained crosstalk values are -12.0 dB and -11.8 dB for the cross and bar state respectively. For *AX06-030* these values are -17.4 dB and -17.2 dB. These last values compare very well to those of switches reported in literature, which have crosstalk values ranging from -12 dB [37] to -25 dB [38].

The reason for the difference in the crosstalk values of the two wafers is due to the mistake made during the processing of *AX06-029*, where the deep etch step had to be redone. The slight mismatch between the two lithographies introduces a non-ideal profile for the MMI, degrading its performance and enhancing the crosstalk.

The Franz-Keldysh effect increases the crosstalk when higher voltages are required for switching, as the optical powers in the two branches of the MZI become different. The measured crosstalk values could therefore be decreased slightly by correcting for the $\delta\varphi$ offset, for example by slightly biasing the phase shifter in the other branch. This lowers the required reverse bias voltages to 0 and V_π. For the same reason, the cross state always shows a lower crosstalk than the bar state.

Losses

For a loss estimation, Fabry-Perot measurements on waveguides that were processed together with the switches were done, as described in section 4.4. These indicated very high losses for both wafers: 17 dB/cm for shallowly etched waveguides and 28 dB/cm for waveguides with metal contacts.

The total loss for the MZI switch is estimated to be 8 dB (3 mm phase shifters) + 3 dB (2 mm waveguides and MMI's) + 1 dB (MMI loss) = 12 dB, which is obviously too high for practical applications. By solving the waveguide loss issue, the total loss should not exceed 5 dB which is acceptable.

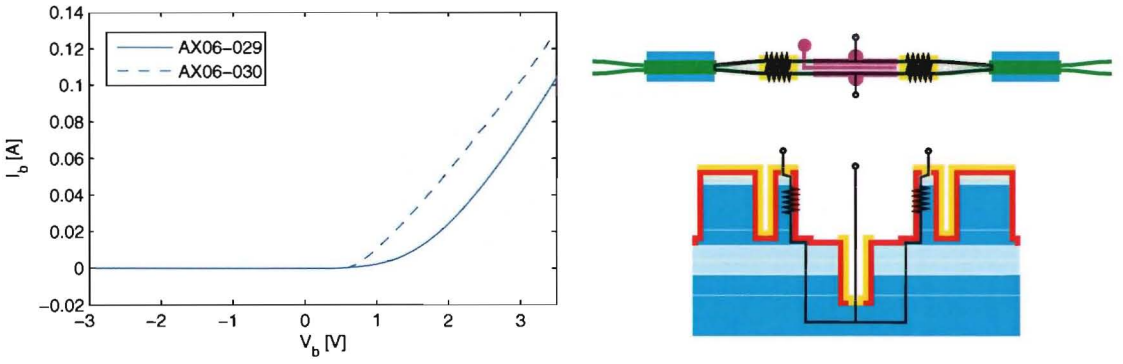


FIGURE 6.7: *Diode behavior of a switch realized on AX06-029 and AX06-030, for reverse and forward bias (a) and possible paths for a leakage current (b).*

WAVELENGTH DEPENDENCE

The switching behavior was checked for wavelengths in the range 1530 to 1560 nm, covering most of the C-band. For both samples, there is only a minor decrease in V_π of about 0.2 V for lower wavelengths. This is to be expected because the Kerr effect increases, the plasma effect decreases and the combined bandfilling and bandgap shrinkage effects increase in strength at higher photon energies.

Furthermore, the curve offset shifts for different wavelengths, with a maximum of 0.6 V difference between $\lambda = 1530$ nm and $\lambda = 1560$ nm, indicating that the MZI is also slightly wavelength dependent because of the imbalance of the two arms. Some variation in the crosstalk is also seen at different wavelengths, with a maximum of 2.5 dB for *AX06-029* and 1 dB for *AX06-030*. For the former this is primarily due to the fabrication imperfections of the MMI, where for the latter it is the absorption (Franz-Keldysh) effect that starts to play a role at lower biases for photon energies closer to the bandgap.

In our design only one polarization has to be considered, making the effective wavelength-independent switching window larger than in designs that have to consider two polarizations. Normally, the strength of the electro-optical effects for TE and TM only overlap at a certain wavelength. Aditionally, the behavior of the MMI has a polarization dependency. This difference in behavior for both polarizations effectively narrows the wavelength independent switching window.

ELECTRICAL BEHAVIOR

In figure 6.7(a) plots of the current-voltage behavior of switches realized on *AX06-029* and *AX06-030* are given, where a positive V_b denotes a forward bias. For low forward bias, the so-called diffusion resistance dominates, while for higher biases the series resistance starts to take over. This bias dependent diffusion resistance is higher for *AX06-029* due to the lower doping at the p-n junction and the transition to a dominating series resistance takes place less abruptly. From figure 6.7(a) the series resistance is seen to be $\frac{\Delta I_b}{\Delta V_b} \approx 20\ \Omega$ for both junctions.

The resistance R of a semiconductor layer with thickness d and surface area A is given by

$$R = \frac{\rho d}{A} \tag{6.2}$$

where the resistivity ρ is given by:

$$\rho = \frac{1}{e\mu\chi} \tag{6.3}$$

$$\tag{6.4}$$

and is either that of electrons in an n-doped material or holes in a p-doped material. Here e is the electron charge, μ is either the hole or electron mobility in a p- or n-doped material respectively and χ is the doping concentration.

Using these equations, the resistance of the layer stack for a 3 μm by 3 mm crossection is determined to be about 10 Ω. The additional 10 Ω is due to the lateral position of the n-contact and the contact resistance.

Both samples showed quite high leakage currents of 100 μA at 10 V reverse bias. In [36] it is pointed out that this is due to Zn passivation of the the p-InP by H_2 during the RIE etch. This creates a small neutralized layer at the walls, through which a current can leak. In the bottom of figure 6.7(b) this electrical path is illustrated, going from the n-contact to the p-contact.

The resistance between two p-contacts was measured to be in the order of 200 kΩ. This number indicates that there is still a connection between both phase shifters, even though most of the top cladding was removed in the thin clad regions. The resistance through the top cladding, illustrated in the top of figure 6.7(b), is calculated to be in the order of 1 MΩ. It is therefore most likely that the current follows the path through the small passivated layer and reaches the other phase shifter through the n-doped substrate (bottom of figure 6.7(b), from p-contact to p-contact). If this was not the case, the switching behavior would be compromised as this would induce a voltage drop over the other phase shifter.

Switching speed

Unfortunately, the behavior under high switching speeds could not be measured as suitable equipment was not available. An educated guess can be made however, as the maximum switching speed is limited by the RC time of the depletion region. Because of the separation of charges under reverse bias, this region can be seen as a capacitor with capacitance C, that is charged or decharged upon switching via a resistor with resistance R. The corresponding RC time is given by:

$$t_{RC} = RC \tag{6.5}$$

where t_{RC} is the time required to either charge the capacitor to $1 - 1/e$ or decharge it to $1/e$ of its maximum value, R is the total resistance and C the capacitance of the depleted region. For a modulation depth of 99% (20 dB), the switching time is $t_s = 4.6t_{RC} = 4.6RC$.

The capacitance of a depleted layer can be approximated by the formula for the capacitance of a parallel plate capacitor with spacing w_d:

$$C = \frac{\epsilon A}{w_d} \tag{6.6}$$

where ϵ is the permittivity of the material, A is the surface area and w_d the depletion width.

We will be switching from zero bias, so that for *AX06-029* $w_d \approx 700$ nm, and for *AX06-030* $w_d \approx 300$ nm (see figures 5.8(a) and 5.8(b)). Assuming a surface of 9000 μm^{-2} (3 mm phase shifting length by 3 μm width), and using equation 6.6, we find $C \approx 1.5$ pF for *AX06-029* and $C \approx 3.5$ pF for *AX06-030*.

With a resistance of 20 Ω, the resulting RC time is now $t_{RC} \approx 30$ ps for *AX06-029* and $t_{RC} \approx 70$ ps for *AX06-030*. This limits the 20 dB switching frequency to 7.2 GHz and 3.1 GHz, respectively. It should be noted that these numbers are only given as a rough indication as other effects, for example the parasitic capacitance of the contact pads, are neglected. It does however indicate that when we want to use carrier effects for electro-optical switching, the maximum operating speed will be significantly influenced.

6.6. Conclusions

We successfully realized efficient switches on both POLIS wafers. Switching can be obtained with a reverse bias of 4.6 V and 2.8 V for a switch with a 3 mm phase shifter realized on *AX06-029* and *AX06-030* respectively. These values correspond to a phase shifting efficiency of 13.0 and 21.4 °/Vmm at 1550 nm. The small discrepancy between the measured and predicted efficiencies is most likely due to the modeled magnitude of the Kerr effect. The measured crosstalk is -12 dB and -17 dB respectively. This difference is due to a small production error during the processing of *AX06-029*. Both samples show a low wavelength dependency, making wavelength independent switching in the C-band range possible. The expected maximum switching speed is in the order of a few GHz, and will be significantly lower for *AX06-030* than for *AX06-029*.

7

Conclusions

7.1. Summary

In this report, we described the realization of Mach-Zehnder based electro-optical switches within the Polarization based Integration Scheme. To come this far, wafers with the right properties were realized based on measurements and quantum well energy level simulations. The relevant electro-optical effects in these materials were determined and quantitively described. Together with a model of the behavior of the electric field in a layer stack and the distribution of the optical field in a phase shifting section, the magnitude of total refractive index change could be predicted. On the basis of this prediction, MZI switches with 3 mm phase shifting sections were realized by some adaptions to an existing design. The switches operate on TM polarized light and show efficient switching behavior.

Based on the simulations, one of the realized wafers had an adjusted layer stack where the quantum well was displaced from the center and the waveguide was slightly doped. From calculations it was predicted to have a higher loss for TM, which was verified by measurements. The doping positively influenced the switching efficiency, making the required switching voltage almost twice as small for this design.

7.2. Evaluation

The POLIS principle and material allow for a straightforward integration of an electro-optical switch. With this, we have come one step further in showing that the POLIS concept is viable. The switches realized require a relatively low driving voltage, and have a low power dissipation because of their reverse bias operation. They have the potential to be used for external laser modulation at high speeds where modulation depth is less of an issue, and for routing purposes at lower speeds with a higher extinction. The fact that only one polarization has to be considered allows for more flexibility in the switch design and increases the wavelength independent behavior.

7.3. Recommendations

Because the additional loss due to a displaced quantum well is rather high, it would be wise to return to having the quantum well in the middle of the waveguide. The benefit from the slight n-doping of the waveguide layer under the quantum well has been proven, so it will still be beneficial to do so in future material designs. A slight increase in switching efficiency over that of *AX06-029* can be expected in this case. Simulations predict a decrease of the required switching voltage of approximately 1 V.

The model as developed in this report has proven to be quite accurate, and the discrepancy can be resolved by reevaluating the magnitude of the Kerr effect. The high required accuracy of the Kramers-Kronig transform and electric field distribution calculation make it very computationally demanding. Thus the model could be improved further to make it more efficient.

An existing switch design was used, which was designed to be polarization independent. Because our switch does not have to meet this demand, a more optimal switch can be designed specifically for TM polarized light. The processing could also be simplified by changing the design so that there is no need for shallowly etched regions, making all structures deeply etched.

All processing steps are standard and allow for easy integration with other devices. The only exception is the removal of the top cladding for electrical isolation. A solution where this step is not necessary to produce the switch would be very welcome.

The maximum switching speed was determined by a simple calculation, high frequency measurements could not be done. To get a valid idea about the actual attainable switching speed, these measurements should be performed. The speed could be increased by operating the switch in gated operation, using both phase shifters and creating a small switching window. If even higher speeds are required, the design could be changed to implement a traveling-wave modulator.

The switches show a rather high leakage current, most likely due to a processing step. The issue of Zn passivation during the RIE etch that causes this leakage, can be solved by an annealing step after etching [36]. The problem of the material losses should also be solved to minimize the total device loss.

APPENDICES

A. CONFINEMENT ENERGY CALCULATIONS

In this appendix, the problem of a particle trapped in a well with finite potential barriers, which is representative for the situation of a carrier in a quantum well, is addressed. Consider figure 1. From quantum mechanics we know that a carrier trapped in region II will have a certain confinement energy and will penetrate into the finite barriers.

For all regions, the time-independent Schrödinger equation in one direction can be written as:

$$\frac{-\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x) \tag{1}$$

where $\hbar$ is the reduced Planck's constant, m the mass of the particle, $\psi(x)$ the wave function (known as an eigenfunction), $U(x)$ the potential function and E the total energy (known as an eigenvalue). Using the information about $U(x)$ from figure 1 ($U = 0$ inside the well, and $U = U_0$ outside the well) we can rewrite the equation for the three regions as follows:

$$\frac{d^2\psi_I(x)}{dx^2} + \alpha^2\psi_I(x) = 0 \tag{2}$$

$$\frac{d^2\psi_{II}(x)}{dx^2} + \kappa^2\psi_{II}(x) = 0 \tag{3}$$

$$\frac{d^2\psi_{III}(x)}{dx^2} + \alpha^2\psi_{III}(x) = 0 \tag{4}$$

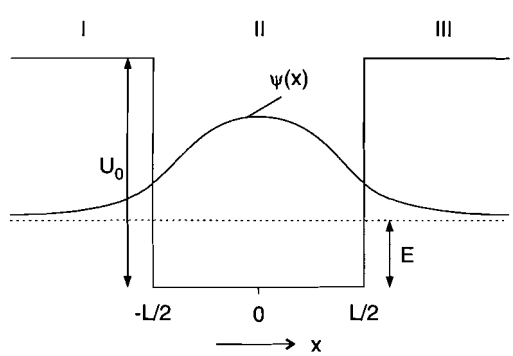


FIGURE 1: *Wavefunction of a particle trapped in a potential well.*

where

$$\alpha = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}} \tag{5}$$

$$\kappa = \sqrt{\frac{2mE}{\hbar^2}} \tag{6}$$

We find the following ground state solution for the wave functions in the three regions:

$$\psi_I(x) = Ae^{\alpha x} \tag{7}$$

$$\psi_{II}(x) = B\cos(\kappa x) \tag{8}$$

$$\psi_{III}(x) = Ce^{-\alpha x} \tag{9}$$

where A, B and C are constants.

The standard constraints on the wave function require that both the wave function and its derivative be continuous at any boundary to be physically meaningful. Thus:

$$\psi_I(-L/2) = \psi_{II}(-L/2) \tag{10}$$

$$\frac{d\psi_I(x)}{dx}|_{x=-L/2} = \frac{d\psi_{II}(x)}{dx}|_{x=-L/2} \tag{11}$$

$$\psi_{III}(L/2) = \psi_{II}(L/2) \tag{12}$$

$$\frac{d\psi_{III}(x)}{dx}|_{x=L/2} = \frac{d\psi_{II}(x)}{dx}|_{x=L/2} \tag{13}$$

Together with the solution for the wave functions, we can write this as:

$$Ae^{-\alpha L/2} = B\cos(-\kappa L/2) \tag{14}$$

$$\alpha Ae^{-\alpha L/2} = -\kappa B\sin(-\kappa L/2) \tag{15}$$

$$Ce^{-\alpha L/2} = B\cos(\kappa L/2) \tag{16}$$

$$\alpha Ce^{-\alpha L/2} = -\kappa B\sin(\kappa L/2) \tag{17}$$

which leads to

$$A = C \tag{18}$$

$$\alpha = \kappa\tan(\kappa L/2) \tag{19}$$

From this equation it follows that the ground state solution of the confinement energy E of a particle with mass m, trapped in a well with width L and barrier height U_0 can be found by solving:

$$\sqrt{\frac{2m(U_0 - E)}{\hbar^2}} = \sqrt{\frac{2mE}{\hbar^2}}\tan\left(\sqrt{\frac{2mE}{\hbar^2}}\frac{L}{2}\right) \tag{20}$$

In a quantum well, the confinement is in one (the perpendicular) direction. Therefore m should be replaced by $m_{lh\perp}$, $m_{hh\perp}$ and $m_{e\perp}$ when calculating the confinement energies of light holes, heavy holes and electrons respectively.

The barrier height for the various particles follows from the calculations of the band gap of a material under strain and is calculated as follows:

$$U_{0lh} = Q_v(E_G - E_g - \Delta E_{\text{bulk}}) - \Delta E_{\text{split}} \tag{21}$$

$$U_{0hh} = Q_v(E_G - E_g - \Delta E_{\text{bulk}}) + \Delta E_{\text{split}} \tag{22}$$

$$U_{0e} = Q_c(E_G - E_g - \Delta E_{\text{bulk}}) \tag{23}$$

where E_G is the bandgap of the surrounding material, E_g the bulk band gap of the quantum well material, ΔE_{bulk} the change in bulk band gap of the quantum well due to strain, ΔE_{split} the hole splitting energy due to strain and Q_v and Q_c are the valence and conduction band offset respectively.

B. Dispersion relations

In this appendix we consider the effective index method to solve the optical field distribution in a three layer slab waveguide in one dimension. This model applies to situation where the high-index waveguide layer is surrounded by lower index cladding layers. The assumption is made that these cladding layers are infinitely thick. Furthermore, it only deals with the real part of refractive indices. In real life these two simplifications imply that the model is only accurate for structures with a thick uniform cladding and low losses.

The optical field in a slab waveguide can be considered to be a standing wave pattern (sine-like) in the direction perpendicular to the interfaces and traveling along the waveguide parallel to the interfaces. The possible realizations of the standing wave pattern are called the modes of the waveguide. The refractive index change at the interface, the thickness of the waveguide and the polarization determine which modes can be supported.

The solutions can be interpreted as a transverse resonance condition: the wave must interfere constructively with itself. This phase matching condition can be written as:

$$k_x d - \phi_0 - \phi_2 = m\pi \tag{24}$$

where k_x is the wave vector in the x-direction, d is the thickness of the waveguide, ϕ_0 and ϕ_2 are the phase shifts at the boundaries of the waveguide material with the top and bottom material respectively and m is the mode number. k_x is given by:

$$k_x = n_1 k_0 \sin\theta \tag{25}$$

where n_1 is the refractive index of the waveguide material, $k_0 = \frac{2\pi}{\lambda_0}$ is the wave vector in a vacuum and θ is the propagation angle in the waveguide.

From the Maxwell equations and boundary conditions at the material interfaces, we can determine the reflection coefficients and from this deduct the phase shifts that were previously mentioned. These shift are different for TE and TM.

$$\phi_{i\text{TE}} = \arctan\left(\frac{\sqrt{n_1^2\cos^2\theta - n_i^2}}{n_1\sin\theta}\right) \tag{26}$$

$$\phi_{i\text{TM}} = \arctan\left(\frac{n_1^2\sqrt{n_1^2\cos^2\theta - n_i^2}}{n_i^2 n_1\sin\theta}\right) \tag{27}$$

where i indicates either the top (0) or bottom (2) interface and n_i is the refractive index of the respective materials. The total dispersion relation can therefore be written as

$$n_1 k_0 \sin\theta d - \arctan\left(\frac{\sqrt{n_1^2\cos^2\theta - n_0^2}}{n_1\sin\theta}\right) - \arctan\left(\frac{\sqrt{n_1^2\cos^2\theta - n_2^2}}{n_1\sin\theta}\right) = m\pi \tag{28}$$

for TE and

$$n_1 k_0 \sin\theta d - \arctan\left(\frac{n_1^2\sqrt{n_1^2\cos^2\theta - n_0^2}}{n_0^2 n_1\sin\theta}\right) - \arctan\left(\frac{n_1^2\sqrt{n_1^2\cos^2\theta - n_2^2}}{n_2^2 n_1\sin\theta}\right) = m\pi \tag{29}$$

for TM.

These equations hold for angles θ below the critical angle:

$$\theta_c = \min\left(\arccos\left(\frac{n_i}{n_1}\right)\right) \tag{30}$$

with $i = 0, 2$. For $\theta < \theta_c$, the wave experiences total reflection and the wave can propagate through the waveguide. For propagation angles larger than the critical angle, the corresponding mode will not be able to propagate.

The values for θ have to be found numerically, from these we can calculate the effective index N_m for a certain mode m:

$$N_m = n_1\cos\theta_m \tag{31}$$

The effective penetration depth of the mode equals

$$d_{m\text{pen}} = \frac{1}{k_0\sqrt{n_1^2\cos\theta_m - n_i^2}} \tag{32}$$

and therefore the effective thickness of the waveguide equals

$$d_{m\text{eff}} = d + \frac{1}{k_0\sqrt{n_1^2\cos\theta_m - n_0^2}} + \frac{1}{k_0\sqrt{n_1^2\cos\theta_m - n_2^2}} \tag{33}$$

We can approximate the distribution of the propagating field of the fundamental mode in the waveguide by a gaussian with a beam waist $w_0 = \frac{d_{m\text{eff}}}{\sqrt{2\pi}}$ and normalization for unity power:

$$U(x) = \sqrt{\frac{2}{d_{m\text{eff}}}}e^{-\frac{2\pi x^2}{d_{m\text{eff}}^2}} \tag{34}$$

where in the case of TE polarization $U(x)$ stands for $E(x)$ (the electric field amplitude) and for TM it is $H(x)$ (its magnetic counterpart). The normalized intensity distribution can then be written as:

$$I(x) = \frac{2}{d_{m\text{eff}}}e^{-\frac{4\pi x^2}{d_{m\text{eff}}^2}} \tag{35}$$

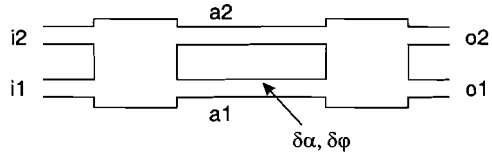


FIGURE 2: *Schematic view of an MZI.*

C. Behavior of an MZI with unbalanced arms

In this appendix we model the behavior of an MZI with unbalance in the arms. Consider the 2x2 3 dB MMI with an input field with intensity P_{in} at port 1 in figure 2. The amplitudes are given by:

$$A_{i1} = \sqrt{P_{in}} \tag{36}$$
$$A_{i2} = 0 \tag{37}$$

This optical power is split into two equal parts, with a relative phase difference of $\frac{\pi}{2}$. The amplitude in both arms can be described by:

$$A_{a1} = \frac{\sqrt{P_{in}}}{\sqrt{2}} \tag{38}$$
$$A_{a2} = \frac{\sqrt{P_{in}}}{\sqrt{2}} e^{-i\frac{\pi}{2}} \tag{39}$$

Assume now that these fields enter a similar MMI. For output port 1, the field from arm 2 gains an extra $\frac{\pi}{2}$ relative phase shift. For output port 2, the field from arm 1 gains an extra $\frac{\pi}{2}$ giving a relative phase difference between both arms of 0. As in the case for only one input field entering the MMI, the power from one arm will split itself among the two output ports giving an additional factor $\frac{1}{\sqrt{2}}$ for the amplitude. We can then simply add the amplitude components at the output ports:

$$A_{o1} = \frac{\sqrt{P_{in}}}{2} + \frac{\sqrt{P_{in}}}{2} e^{-i\pi} = 0 \tag{40}$$
$$A_{o2} = \frac{\sqrt{P_{in}}}{2} + \frac{\sqrt{P_{in}}}{2} = \sqrt{P_{in}} \tag{41}$$

and for the output powers:

$$P_{o1} = A_{o1} A_{o1}^* = 0 \tag{42}$$
$$P_{o2} = A_{o2} A_{o2}^* = P_{in} \tag{43}$$

Now we introduce an imbalance by adding an additional relative phase shift $\delta\varphi$ and (intensity) attenuation $\delta\alpha$ in arm 1.

$$A_{a1} = \frac{\sqrt{P_{in}}}{\sqrt{2}} \sqrt{1 - \delta\alpha} e^{-i\delta\varphi} \tag{44}$$
$$A_{a2} = \frac{\sqrt{P_{in}}}{\sqrt{2}} e^{-i\frac{\pi}{2}} \tag{45}$$

Combining these amplitudes at the output ports under the conditions mentioned before, gives:

$$A_{o1} = \frac{\sqrt{P_{in}}}{2}\sqrt{1-\delta\alpha}e^{-i\delta\varphi} + \frac{\sqrt{P_{in}}}{2}e^{-i\pi} = \frac{\sqrt{P_{in}}}{2}\left(\sqrt{1-\delta\alpha}e^{-i\delta\varphi} - 1\right) \tag{46}$$

$$A_{o1} = \frac{\sqrt{P_{in}}}{2}\sqrt{1-\delta\alpha}e^{-i\delta\varphi} + \frac{\sqrt{P_{in}}}{2} = \frac{\sqrt{P_{in}}}{2}\left(\sqrt{1-\delta\alpha}e^{-i\delta\varphi} + 1\right) \tag{47}$$

and for the output powers:

$$P_{o1} = A_{o1}A_{o1}^* = \frac{P_{in}}{4}\left(2 - \delta\alpha - 2\sqrt{1-\delta\alpha}\cos\delta\varphi\right) \tag{48}$$

$$P_{o2} = A_{o2}A_{o2}^* = \frac{P_{in}}{4}\left(2 - \delta\alpha + 2\sqrt{1-\delta\alpha}\cos\delta\varphi\right) \tag{49}$$

These equations obviously reduce to those of 40 and 41 for $\delta\alpha = 0$ and $\delta\varphi = 0$.

This imbalance can have an intended (e.g. applying a voltage on one arm) or unintended (production errors) origin. In the first case it can be used to obtain switching behavior, whereas the latter is usually unwanted. In this derivation, losses and crosstalk are not taken into account.

D. InGaAsP material parameters

Parameters used for quantum well calculations

Material parameters for $In_xGa_{1-x}As_yP_{1-y}$, used for e-hh and e-lh transition energy calculations. The parameters are defined in section 2.2 and appendix A. The values are taken from reference [12], Q_c is taken from reference [13]:

a_0	Å	$5.6533(1-x)y + 6.0583xy + 5.4505(1-x)(1-y) + 5.8687x(1-y)$
c_{11}	10^5 bar	$11.9(1-x)y + 8.329xy + 14.05(1-x)(1-y) + 10.11x(1-y)$
c_{12}	10^5 bar	$5.38(1-x)y + 4.526xy + 6.203(1-x)(1-y) + 5.61x(1-y)$
E_g	eV	$1.35 - 1.17y + 0.668(1-x) - 0.069y(1-x) + 0.18y^2 + 0.03y^2(1-x) + 0.785(1-x)^2 - 0.322y(1-x)^2$
dE_g/dP	10^{-5} eV/bar	$1.13(1-x)y + 1.02xy + 1.07(1-x)(1-y) + 0.84x(1-y)$
a	eV	$-\frac{1}{3}dE_g/dP(c_{11} + 2c_{12})$
b	eV	$-1.7(1-x)y - 1.8xy - 1.8(1-x)(1-y) - 2.0x(1-y)$
m_e	m_0	$0.0632(1-x)y + 0.0213xy + 0.158(1-x)(1-y) + 0.077x(1-y)$
$m_{lh\perp}$	m_0	$0.088(1-x)y + 0.024xy + 0.16(1-x)(1-y) + 0.12x(1-y)$
$m_{hh\perp}$	m_0	$0.5(1-x)y + 0.5172xy + 0.54(1-x)(1-y) + 0.56x(1-y)$
$m_{lh\parallel}$ [a]	m_0	$0.23(1-x)y + 0.084xy + 0.34(1-x)(1-y) + 0.2921x(1-y)$
$m_{hh\parallel}$ [b]	m_0	$0.11(1-x)y + 0.0315xy + 0.19(1-x)(1-y) + 0.1493x(1-y)$
Q_c	-	0.43

[a] Under strain, otherwise $m_{lh\parallel} = m_{lh\perp}$

[b] Under strain, otherwise $m_{hh\parallel} = m_{hh\perp}$

General parameters of InGaAsP

Except for the quantum well calculations, we are dealing with materials lattice-matched to InP. This matching condition relates x and y in $In_xGa_{1-x}As_yP_{1-y}$ by:

$$x = \frac{16.17y - 33.41}{y - 33.41} \tag{50}$$

Because extensive experimental values have been measured for these materials, we will use these wherever possible. The following parameters are taken from reference [14]:

E_g	eV	$1.35 - 0.72y + 0.12y^2$
m_e	m_0	$0.07 - 0.0308y$
m_{lh}	m_0	$0.12 - 0.078y + 0.002y^2$
m_{hh}	m_0	$0.6 - 0.218y + 0.07y^2$
ϵ (static)	ϵ_0	$12.35 + 1.62y - 0.055y^2$

For the refractive index for photon energies at least 0.2 eV below the bandgap energy, we adopt the modified Sellmeier formula:

$$n = \sqrt{A + \frac{B\lambda_0^2}{\lambda_0^2 - C}} \tag{51}$$

where λ_0 [μm] is the wavelength in a vacuum and the Sellmeier coefficients for $InGaAs_yP_{1-y}$ lattice matched to InP are given by [14]:

A	$7.255 + 1.15y + 0.489y^2$
B	$2.316 + 0.604y - 0.493y^2$
C	$0.3922 + 0.396y + 0.158y^2$

The refractive index for photon energies 0 to 0.2 eV below the bandgap energy is given by [15]:

$$n = 3.425 + 0.940\Delta E + 0.952(\Delta E)^2 + (0.255 - 0.257\Delta E)y - (0.103 - 0.0952\Delta E)y^2 \tag{52}$$

where y is the arsenic fraction of $InGaAs_yP_{1-y}$ and $\Delta E = E_{ph} - E_g$ [eV].

For the carrier mobility in p- and n-type semiconductors, reference [18] gives the empirical formula and fitted parameters. We use these values to arrive at the following expressions for the binary compounds, where P [cm^{-3}] and N [cm^{-3}] are the doping levels for p- and n-doped material respectively:

Material	μ_h [$cm^2V^{-1}s^{-1}$]	μ_e [$cm^2V^{-1}s^{-1}$]
InAs	$20 + \frac{3.52\cdot10^{10}}{P^{0.46}+6.90\cdot10^6}$	$1000 + \frac{1.96\cdot10^{10}}{N^{0.32}+5.93\cdot10^5}$
InP	$10 + \frac{6.17\cdot10^{13}}{P^{0.62}+3.86\cdot10^{11}}$	$\frac{4.17\cdot10^{11}}{N^{0.45}+8.34\cdot10^7}$
GaAs	$20 + \frac{1.58\cdot10^9}{P^{0.38}+3.35\cdot10^6}$	$500 + \frac{3.63\cdot10^{10}}{N^{0.394}+4.08\cdot10^6}$
GaP	$10 + \frac{2.73\cdot10^{17}}{P^{0.85}+2.00\cdot10^{15}}$	$10 + \frac{1.17\cdot10^{17}}{N^{0.8}+8.22\cdot10^{14}}$

The value for $In_xGa_{1-x}As_yP_{1-y}$ can be calculated using Vegard's law:

$$\mu_{InGaAsP} = xy\mu_{InAs} + x(1-y)\mu_{InP} + (1-x)y\mu_{GaAs} + (1-x)(1-y)\mu_{GaP} \tag{53}$$

E. TM E-FIELD DISTRIBUTION

In this appendix we take a closer look at the field distributions of a TM mode traveling through a slab waveguide. We start with a traveling wave in the z-direction, for which the electric field **E** has the following form:

$$\mathbf{E}(z,t) = E_0 e^{i(\omega t - \beta z)} \tag{54}$$

where E_0 is the electric field amplitude and a similar relation can be written for **H**. ω denotes the angular frequency of the traveling wave and β the propagation constant of the mode in the z-direction.

For an isotropic, non-magnetic and non-dispersive material, we can write Maxwell's equations in the following simple form:

$$\nabla \times \mathbf{H} = \epsilon_0 n^2 \frac{\partial \mathbf{E}}{\partial t} \tag{55}$$

$$\nabla \times \mathbf{E} = \mu_0 \frac{\partial \mathbf{H}}{\partial t} \tag{56}$$

where ϵ_0 and μ_0 are the permittivity and permeability of a vacuum respectively.

A TM mode traveling through a slab waveguide as depicted in figure 4.5(b) has the field components H_y, E_x and E_z. Expanding Maxwell's equations 55 and 56 and setting $H_x = H_z = E_y = 0$ and $\frac{\partial}{\partial y} = 0$ for all field components (as we are considering a slab waveguide with no variation in the y direction), we produce the following set of equations:

$$-\frac{\partial H_y}{\partial z} = \epsilon_0 n^2 \frac{\partial E_x}{\partial t} \tag{57}$$

$$\frac{\partial H_y}{\partial x} = \epsilon_0 n^2 \frac{\partial E_z}{\partial t} \tag{58}$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\mu_0 \frac{\partial H_y}{\partial t} \tag{59}$$

E and **H** have the same z-dependence given in 54. We can write for the electric field:

$$\frac{\partial \mathbf{E}}{\partial t} = E_0 \frac{\partial e^{i(\omega t - \beta z)}}{\partial t} = E_0 i\omega e^{i(\omega t - \beta z)} = i\omega \mathbf{E} \tag{60}$$

$$\frac{\partial \mathbf{E}}{\partial z} = E_0 \frac{\partial e^{i(\omega t - \beta z)}}{\partial t} = -E_0 i\beta e^{i(\omega t - \beta z)} = -i\beta \mathbf{E} \tag{61}$$

and a similar set of equations can be written for the magnetic field.

Using these equations, we can write for 57 to 59:

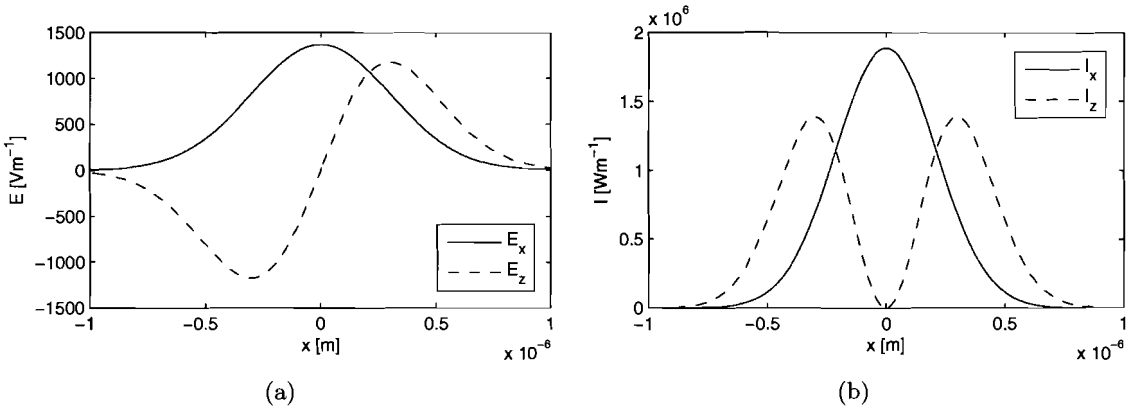
$$i\beta H_y = i\omega \epsilon_0 n^2 E_x \tag{62}$$

$$\frac{\partial H_y}{\partial x} = i\omega \epsilon_0 n^2 E_z \tag{63}$$

$$i\beta E_x + \frac{\partial E_z}{\partial x} = i\omega \mu_0 H_y \tag{64}$$

Combining these equations we arrive at the one-dimensional wave equation for H_y:

$$\frac{\partial^2 H_y}{\partial x^2} + (n^2 k^2 - \beta^2) H_y = 0 \tag{65}$$



(a) (b)

FIGURE 3: *Normalized field (a) and intensity (b) distribution of the electric field components of the fundamental TM mode.*

where we have used $\omega^2\mu_0\epsilon_0 = k^2$ where k is the wavenumber of the propagating mode. The zero-order solution to the equation is a cosine in the waveguide and a decaying exponential in the cladding layers.

It is important to note that the E_x component of the electric field in the x-direction is directly coupled to that of H_y, whereas the E_z component takes the form of the derivative of this distribution and is therefore sine-like as a function of x. It will be zero in the middle of a symmetric waveguide (where E_x is maximum).

E_x can be approximated by a gaussian, as given in appendix B:

$$E_x(x) = \sqrt{\frac{2}{d_{meff}}} e^{-\frac{2\pi x^2}{d_{meff}^2}} \tag{66}$$

whereas E_z takes the form of its derivative, and can be written as:

$$E_z(x) = \sqrt{\frac{16\pi}{d_{meff}^3}} x e^{-\frac{2\pi x^2}{d_{meff}^2}} \tag{67}$$

both normalized to unity power[1]. The normalized intensities are then given by:

$$I_x(x) = \frac{2}{d_{meff}} e^{-\frac{4\pi x^2}{d_{meff}^2}} \tag{68}$$

$$I_z(x) = \frac{16\pi}{d_{meff}^3} x^2 e^{-\frac{4\pi x^2}{d_{meff}^2}} \tag{69}$$

An illustration of these distributions is give in figure 3(a) and 3(b).

The E_z component does not contribute to the flow of power in the direction of propagation (the z-direction), whereas the E_x component does.

[1]This is obviously not the actual power in watts, but a relative number representing it. The same goes for the intensity.

F. Magnitude of the bandfilling effect

This appendix deals with determining the magnitude of p_{abs} as it is used for calculation of the bandfilling effect. As discussed in section 5.3, the relevant probabilities are given as:

$$p_{\text{abs}lh} = f(E_{alh}) - f(E_{blh}) \tag{70}$$

$$p_{\text{abs}hh} = f(E_{ahh}) - f(E_{bhh}) \tag{71}$$

For the transition of an electron under absorption of a photon with energy E_{ph}, both the initial and final state should be allowed in the bands. This relates the states E_a and E_b to the photon energy. Using equations 2.8 and 2.9, having $E_b = E_a + E_{ph}$ (conservation of energy) and solving for k^2 (conservation of momentum), we obtain:

$$E_{alh} = (E_g - E_{ph})\left(\frac{m_e}{m_e + m_{lh}}\right) - E_g \tag{72}$$

$$E_{ahh} = (E_g - E_{ph})\left(\frac{m_e}{m_e + m_{hh}}\right) - E_g \tag{73}$$

$$E_{blh} = (E_{ph} - E_g)\left(\frac{m_{lh}}{m_e + m_{lh}}\right) \tag{74}$$

$$E_{bhh} = (E_{ph} - E_g)\left(\frac{m_{hh}}{m_e + m_{hh}}\right) \tag{75}$$

where the minimum of the conduction band is set to $E = 0$, E_{alh} indicates the initial level involving a light hole transition, etc.. E_g is the bandgap of the material and m_e, m_{lh} and m_{hh} are the electron, light hole and heavy hole effective masses respectively.

$f(E)$ is given by the Fermi-Dirac distribution function:

$$f(E) = \frac{1}{1 + e^{\frac{E - E_F}{kT}}} \tag{76}$$

where E_F is the Fermi-level[2], k is the Boltzmann constant and T is the temperature (300 K).

The approximation for the Fermi-level that can be used depends on the degeneracy. A semiconductor is defined to be degenerate when its Fermi level approaches either conduction or valence band within $3kT \approx 52$ meV. For a non-degenerate semiconductor the Fermi-level can be approximated using the Boltzmann approximation for p- (E_{PF}) or n-type (E_{NF}) doping:

$$E_{PF} = -\ln\left(\frac{P}{N_v}\right) kT - E_g \tag{77}$$

$$E_{NF} = \ln\left(\frac{N}{N_c}\right) kT \tag{78}$$

where the minimum of the conduction band is set to $E = 0$, P and N are the doping levels and N_v and N_c are the effective density of states in valence and conduction bands respectively.

[2]We are only considering the Fermi-level, as we have either p or n doping. In the case of carrier-injection, the quasi Fermi-levels would have to be considered.

The latter are given by:

$$N_v = 2\left(\frac{2\pi m_h kT}{h^2}\right)^{\frac{3}{2}} \tag{79}$$

$$N_c = 2\left(\frac{2\pi m_e kT}{h^2}\right)^{\frac{3}{2}} \tag{80}$$

where m_h is the effective hole mass given by equation 4.5 and m_e is the effective electron mass.

From equations 79 and 80 and the definition of degeneracy, we see that above $P \approx \frac{Nv}{20}$ and $N \approx \frac{Nc}{20}$ the semiconductor is assumed to be degenerate. In this case we use the Joyce-Dixon approximation:

$$E_{PF} = -\left[\ln\left(\frac{P}{N_v}\right) + \frac{1}{\sqrt{8}}\left(\frac{P}{N_v}\right)\right] kT - E_g \tag{81}$$

$$E_{NF} = \left[\ln\left(\frac{N}{N_c}\right) + \frac{1}{\sqrt{8}}\left(\frac{P}{N_c}\right)\right] kT \tag{82}$$

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List of constants and units

c	Speed of light in a vacuum	$3.00 \cdot 10^{8}$ ms^{-1}
e	Elementary electron charge	$1.60 \cdot 10^{-19}$ C
ϵ_0	Permittivity of a vacuum	$8.85 \cdot 10^{-12}$ Fm^{-1}
eV	Electronvolt	$1.60 \cdot 10^{-19}$ J
h	Planck's constant	$6.63 \cdot 10^{-34}$ Js
$\hbar$	Reduced Planck's constant	$1.05 \cdot 10^{-34}$ Js
k	Boltzmann constant	$1.38 \cdot 10^{-23}$JK^{-1}
m_0	Free electron rest mass	$9.11 \cdot 10^{-31}$ kg

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